Neuroscience

Restoration of locomotor function following stimulation of the A13 region in Parkinson's mouse models

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eLife Assessment

This **valuable** study reveals the pro-locomotor effects of activating a deep brain region containing diverse range of neurons in both healthy and Parkinson's disease mouse models. While the findings are **solid**, mechanistic insights remain limited due to the small sample size. This research is relevant to motor control researchers and offers clinical perspectives.

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Abstract

Parkinson's disease (PD) is characterized by extensive motor and non-motor dysfunction, including gait disturbance, which is difficult to treat effectively. This study explores the therapeutic potential of targeting the A13 region, a heterogeneous region of the medial zona incerta (mZI) containing dopaminergic, GABAergic, and glutamatergic neurons that has shown relative preservation in PD models. The A13 is identified to project to the mesencephalic locomotor region (MLR), with a subpopulation of cells displaying activity correlating to movement speed, suggesting its role in locomotion. We show that photoactivation of this A13 region can alleviate bradykinesia and akinetic features, while increasing turning in a mouse model of PD. These effects combine disease-specific rescue of function with a possible gain of function. We identified areas of preservation and plasticity within the mZI connectome using whole-brain imaging. Our findings suggest a global remodeling of afferent and efferent projections of the A13 region, highlighting the zona incerta's role as a crucial hub for the rapid selection of motor function. The study unveils the significant pro-locomotor effects of the A13 region and suggests its promising potential as a therapeutic target for PD-related gait dysfunction.

Significance statement

This work examines the function of the A13 region in locomotion, an area with direct connectivity to locomotor regions in the brainstem. A13 stimulation can restore locomotor function and improve bradykinesia in a PD mouse model.

Introduction

Parkinson's disease (PD) is a complex condition affecting many facets of motor and non-motor functions, including visual, olfactory, memory and executive functions (Cenci and Björklund, 2020). Due to the widespread features of PD, focusing on changes within a single pathway cannot account for all symptoms. Gait disturbance is one of the hardest to treat; pharmacological, deep brain stimulation (DBS) and physical therapies lead to only partial improvements (Nonnekes et al., 2020, 2015). While the subthalamic nucleus (STN) and globus pallidus (GPi) are common DBS targets for PD, alternative targets such as pedunculopontine nucleus (PPN) and the zona incerta (ZI) have been proposed with mixed results in improving postural and/or gait dysfunctions (Caire et al., 2013; Ferraye et al., 2010; Gut and Winn, 2015; Hamani et al., 2011; Moro et al., 2010; Nonnekes et al., 2015; Okun and Foote, 2010; Ossowska, 2019; Stefani et al., 2007; Thevathasan et al., 2018). Part of the issue with targeting the ZI with DBS strategies is the relative lack of knowledge regarding its downstream anatomical and functional connectivity with motor centers. Recent work with photoactivation of subpopulations of PPN neurons in PD models shows promise for similar ZI-focused strategies (Masini and Kiehn, 2022).

The ZI is recognized as an integrative hub, with roles in regulating sensory inflow, arousal, motor function, and conveying motivational states (Chometton et al., 2017; Mitrofanis, 2005; Monosov et al., 2022; Sharma et al., 2024; Wang et al., 2020; Yang et al., 2022; Zhao et al., 2019). As such, it is well placed to be involved in PD and has seen increased clinical and preclinical research over the last two decades (Blomstedt et al., 2018; Ossowska, 2019; Plaha et al., 2008). However, little attention has been placed on the medial zona incerta (mZI), particularly the A13, the only dopamine-containing region of the rostromedial ZI (Bolton et al., 2015; Kim et al., 2017; Sharma et al., 2018). Recent research in primates and mice (Peoples et al., 2012; Roostalu et al., 2019; Shaw et al., 2010) indicates that the A13 is preserved in 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP)-based PD models.

Recently, we discovered that the A13 located within the ZI projects to two areas of the mesencephalic locomotor region (MLR), the PPN and cuneiform nucleus (CUN) (Sharma et al., 2018), suggesting a role for A13 in locomotor function. Indeed, mini-endoscopic calcium recordings from calcium/calmodulin-dependent protein kinase IIα (CaMKIIα) populations in the rostral ZI, which includes the A13 nucleus, show a subpopulation of cells whose activity correlates with movement speed (Z. Li et al., 2021). Since this region projects to the MLR, it is a potential parallel motor pathway to target for gait improvement. Photoactivation of glutamatergic MLR neurons alleviates motor deficits in mouse models that transiently blocked dopamine transmission or lesioned SNc with 6-OHDA (Fougère et al., 2021; Masini and Kiehn, 2022). Phenomena such as kinesia paradoxa (Glickstein and Stein, 1991) in PD patients support the existence of preserved parallel motor pathways that can be engaged in particular circumstances to produce normal movement.

Further evidence supporting the importance of parallel motor pathways in PD includes those reporting functional alterations in A13 (Hoffman et al., 1997; Périer et al., 2000). Nigrostriatal lesions affect A13 cellular function and lead to anatomical remodeling in monoaminergic brain regions including widespread alterations in dopaminergic, noradrenergic, cholinergic, serotoninergic, and other monoaminergic neuronal populations (Braak et al., 2003; Kish et al., 2008; Lim et al., 2009; Perez-Lloret and Barrantes, 2016; Roostalu et al., 2019; Scatton et al., 1983; Zweig et al., 1989). There is additional evidence showing parallel motor pathways in the A13. For example, the A13 connectome encompasses the cerebral cortex (Mitrofanis and Mikuletic, 1999), central nucleus of the amygdala (Eaton et al., 1994), thalamic paraventricular nucleus (Li et al., 2014), thalamic reuniens (Sita et al., 2007; Venkataraman et al., 2021), CUN and PPN (Sharma et al., 2018), superior colliculus (SC) (Bolton et al., 2015), and dorsolateral periaqueductal grey (PAG) (Messanvi et al., 2013; Sita et al., 2007), making the A13 an potential hub for goal-directed locomotion (Choi and McNally, 2017; Eaton et al., 1994; Messanvi et al., 2013; Mok and Mogenson, 1986; Moriya et al., 2020; Ogundele et al., 2017; Manjit K. Sanghera et al., 1991; M. K. Sanghera et al., 1991; Sita et al., 2007; Venkataraman et al., 2021).

Considering the role of the A13 region in gait control, particularly its potential as a therapeutic target to enhance gait in Parkinson's Disease (PD), we focused on exploring the benefits of photoactivating this anatomically defined area. This targeted region predominantly encompasses the A13, along with a portion of the medial zona incerta (mZI), collectively referred to as the A13 region throughout this study. We identified areas of preservation and plasticity within the mZI connectome using whole-brain imaging techniques. While endogenous compensatory mechanisms from the remaining and remodelled A13 region connectome were inadequate in overcoming locomotor deficits, photoactivation of the A13 region rescued bradykinetic and akinetic symptoms in a mouse model of 6-hydroxydopamine (6-OHDA) mediated unilateral nigrostriatal degeneration. These findings indicate that the A13 region produces powerful pro-locomotor effects in normal and PD mouse models. Moreover, PD-related bradykinesia is ameliorated with A13 region photoactivation in the presence of remodelling of the A13 region connectome. Because we found the zona incerta is a hub for the rapid selection of motor function (Wang et al., 2020), we then mapped the input and output patterns finding evidence of global remodeling of afferent and efferent projections of the A13 region. Some of these data have been published in abstract form (L. Kim et al., 2021).

Results

Unilateral 6-OHDA mouse model has robust motor deficits

The overall experimental design is illustrated in **Figure 1A**, along with a schematic in **Figure 1B** showing injections of 6-OHDA in the medial forebrain bundle (mFB) and AAVDJ-CaMKIIα-ChR2 virus into the medial zona incerta (mZI). We confirmed substantia nigra pars compacta (SNc) degeneration in a well-validated unilateral 6-OHDA-mediated Parkinsonian mouse model (Thiele et al., 2012). The percentage of tyrosine hydroxylase (TH^+) cell loss normalized to the intra-animal contralesional side was quantified. 6-OHDA produced a significant lesion that decreased TH^+ neuronal SNc populations. As previously reported (Boix et al., 2015), the SNc ipilesional to the 6-OHDA injection (n = 10) showed major ablation of the TH^+ neurons compared to sham animals (**Figure 1C and D**: n = 12).

A13 region photoactivation generates pro-locomotor behaviors in the open field

6-OHDA lesions produce bradykinetic and akinetic phenotypes in the open field (L.-X. Li et al., 2021; Magno et al., 2019; Sanders and Jaeger, 2016). To understand the impact of A13 region photoactivation on locomotion in sham and PD model mice, on-target localization of the optical fiber above the A13 region, centered on the mZI, along with YFP reporter expression, was

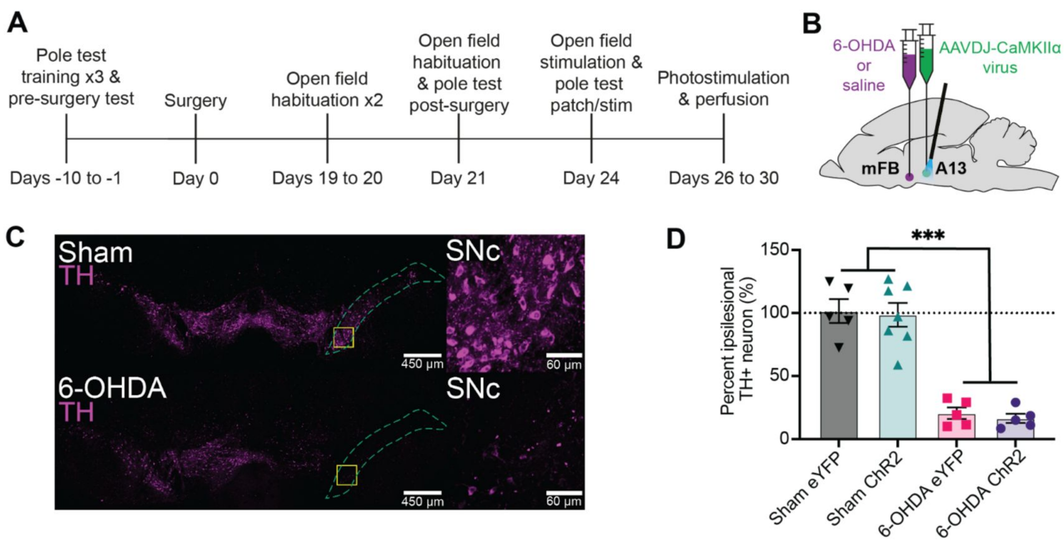


Figure 1.

Experimental design and confirmation of unilateral TH^+ depletion in the SNc via 6-OHDA lesion.

(A) Illustration of experimental timeline. (**B**) Dual ipsilateral stereotaxic injection into the mFB and A13 region. (**C**) TH^+ cells in SNc of sham (top) compared to 6-OHDA injected mouse (bottom). Magnified areas outlined by yellow squares are shown on the right. (**D**) Unilateral injection of 6-OHDA (6-OHDA ChR2: n = 5, 6-OHDA eYFP: n = 5) into the MFB resulted in greater percentage of TH^+ loss compared to sham in the SNc (sham ChR2: n = 7, sham eYFP: n = 5, 2-way ANOVA), regardless of virus type ($F_{1,18}$ = 104.4, p < 0.001). ***p < 0.001. Error bars indicate SEMs.

confirmed (**Figure 2**; **Figure S1**) in mice given sham or 6-OHDA injections. Corroborating the *post hoc* targeting, we found evidence for c-Fos in neurons within the A13 region in photostimulated ChR2 mice (**Figure 2D**). We observed an increase in c-Fos fluorescence intensity in the A13 region of photostimulated 6-OHDA ChR2 mice compared to the 6-OHDA eYFP mice. There was also an increase in the number of ChR2 cells with c-Fos labeling in 6-OHDA ChR2 mice compared to the 6-OHDA eYFP mice. However, there was no net increase in TH^+ cells labeled with ChR2 and c-Fos suggesting a heterogeneous population of activated cells. Before *post hoc* analysis, mice were monitored in the open field test (OFT), where the effects of the 6-OHDA lesion were apparent, with 6-OHDA lesioned animals demonstrating far less movement, fewer bouts of locomotion, and less time engaging in locomotion (**Figure 3A-E**). Using instantaneous animal movement speeds that exceeded 2 cm/s as per Masini & Kiehn (2022), we plotted instantaneous speed (**Figure 3B-E**) and analyzed one-minute bins (**Figure 3H**). As was expected, 6-OHDA lesioned animals had lower movement speeds than sham control animals ($p < 0.001$). One animal from the 6-OHDA eYFP group was excluded because it did not meet the speed threshold during recording. Notably, photoactivation of the A13 region often generated dramatic effects, with mice showing a distinct increase in locomotor behavior (**Figure 3A**, Movie S1 & Movie S2). Both sham and 6-OHDA ChR2 mice showed a significant increase in locomotor distance travelled during periods of photoactivation (**Figure 3F**, $p < 0.01$). One sham animal showed grooming behavior on stimulation and was excluded from the analysis.

We tested whether photoactivation led to a single bout of locomotion or if there was an overall increase in bouts, signifying that animals could repeatedly initiate locomotion following photoactivation. Mice in the ChR2 groups demonstrated an increase in the number of locomotion bouts with photoactivation, indicating a greater ability to start locomotion from rest, and that photoactivation was not eliciting a single prolonged bout (**Figure 3G**, $p < 0.01$). Photoactivation increased the total duration of locomotor bouts (**Figure 3I**, $p < 0.001$). We did note a refractory decrease in the distance travelled by the sham ChR2 animal group. To control for this, we compared the pre-stimulation timepoints to the baseline one-minute averages to ensure that the animal locomotion distance travelled returned to a stable state before stimulation was reapplied (**Figure S2**, $p = 0.78$).

Next, we examined the reliability of photoactivation to initiate locomotion. The percentage of trials with at least one bout of locomotion was compared for the pre- and stim time points. 6-OHDA ChR2 animals showed a reliable pro-locomotion phenotype with A13 region photoactivation (**Figure S3A**, $p < 0.05$). As was expected in the control 6-OHDA eYFP group, there was no effect of photoactivation on the probability of engaging in locomotion (**Figure S3A**, $p = 0.71$).

Animal movement speed also factors into the total distance travelled measure and can be discussed in regard to a bradykinetic phenotype in 6-OHDA lesioned mice (Magno et al., 2019; Sanders and Jaeger, 2016). The 6-OHDA and sham ChR2 groups displayed increases in average speed in comparison to the 6-OHDA and sham eYFP groups during photostimulation (**Figure 3H**, $p < 0.001$). When we examined the time taken to initiate locomotion, there was no significant difference between sham or 6-OHDA ChR2 groups (**Figure S3B**, $p = 0.95$).

Photoactivation of the A13 region increases ipsilesional turning in the open field test

Unilateral 6-OHDA lesions drive asymmetric rotational bias (Boix et al., 2015; L.-X. Li et al., 2021; Magno et al., 2019; Thiele et al., 2012). We were interested in whether this persisted with stimulation and noted upon observation of photoactivation that animals appeared to have increased ipsilesional rotation. As observed, 6-OHDA ChR2 animals had an increase in turn angle sum (TAS), indicating an increase in their rotational bias with photoactivation in the ipsilesional direction (**Figure 3J**, $p < 0.001$). As expected, 6-OHDA eYFP animals showed consistent rotational bias throughout time. The rotational bias of sham ChR2 was also compared to determine whether

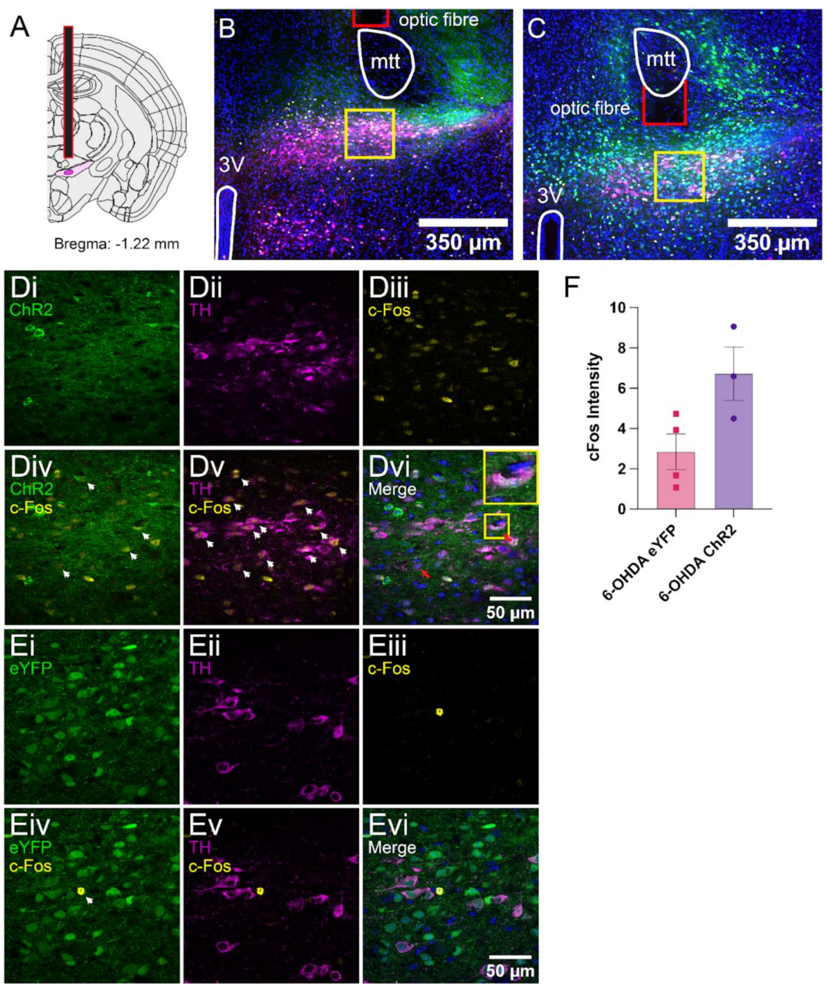


Figure 2.

Post hoc c-Fos expression and targeting of the mZI and A13.

(A) Diagram showing the A13 DAergic nucleus in dark magenta encapsulated by the ZI in light magenta. The fiber optic tip is outlined in red. Atlas image adapted from the Allen Brain Atlas (Goldowitz, 2010). **(B)** Tissue images were obtained from a 6-OHDA ChR2 animal and **(C)** a 6-OHDA eYFP animal. Images show the distribution of DAPI (blue), eYFP (green), c-Fos (yellow), and TH (magenta). Landmarks are outlined in white (3V: third ventricle; A13 and mZI as shown in A), and the optic cannula tip is shown in red. Higher magnification images of the A13 DAergic nucleus are outlined by the yellow boxes in a 6-OHDA ChR2 animal **(Di-Dvi)** and a 6-OHDA eYFP animal **(Ei-Evi)**. Images show isolated channels in the top rows of the respective groups: eYFP **(i)**, TH **(ii)**, and c-Fos **(iii)**. Merged channels for eYFP and c-Fos **(iv)**, TH and c-Fos **(v)**, and a merge of all four channels **(vi)** are presented in the bottom rows of their respective groups. White arrowheads in the merged images highlight overlap in merged markers. Red arrows show triple colocalization of eYFP, c-Fos and TH. **(Dvi)** contains a magnified example of triple-labelled neurons, as highlighted in the yellow box. **(F)** Graph shows increase in the intensity of c-Fos fluorescence after photoactivation in 6-OHDA ChR2 mice (p = 0.05).

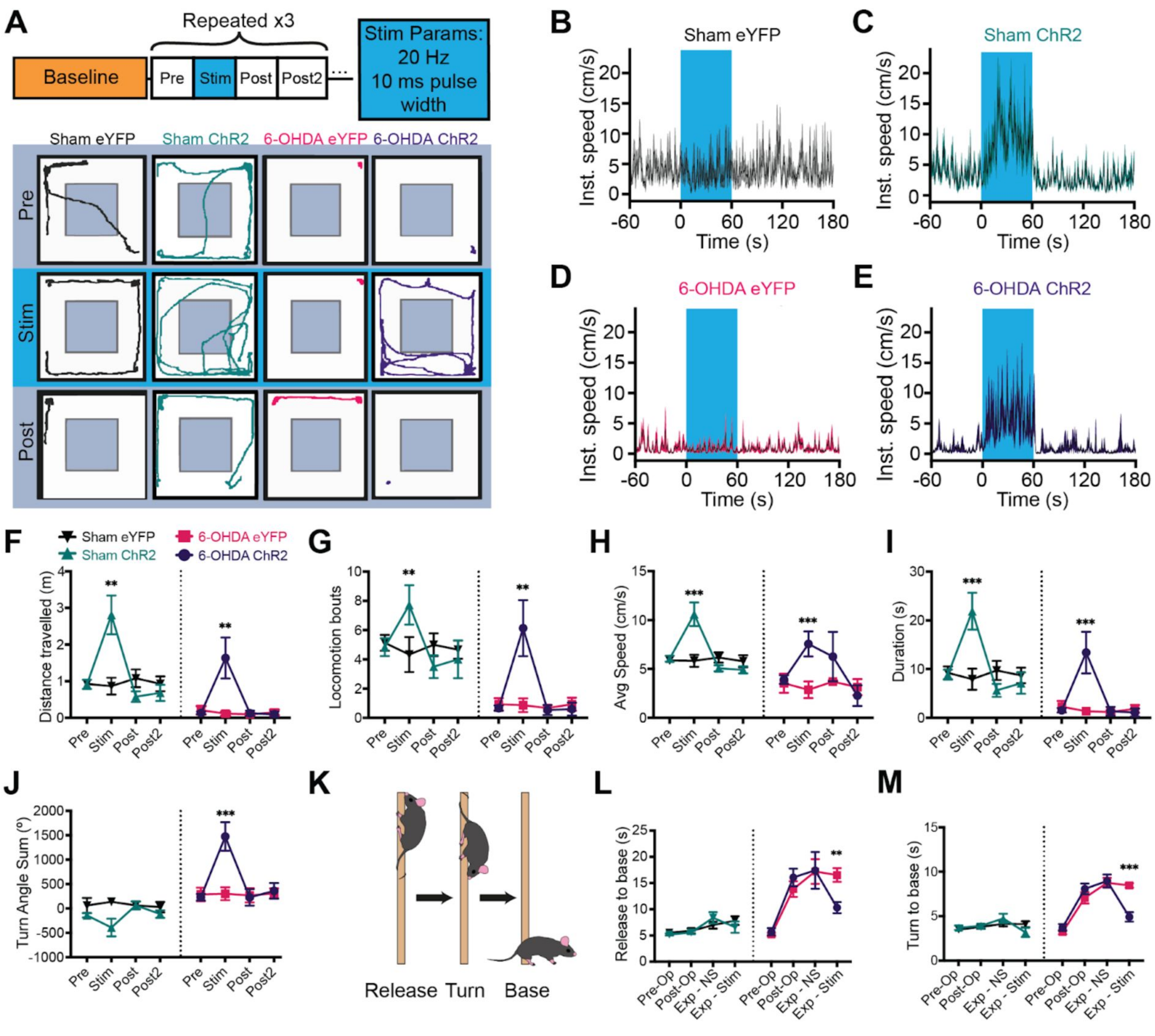


Figure 3.

Ispilesional photoactivation of the A13 region in a unilateral 6-OHDA mouse model rescues motor deficits.

(A) Schematic of open field experiment design and example traces for open field testing (each testing box is 1 min - total test is 4 min) with unilateral photoactivation of the A13 region. (**B-E**) Group averaged instantaneous velocity graphs showing no increase in a sham eYFP (**B**) or 6-OHDA eYFP mouse (**C**), with increases in velocity during stimulation in a sham ChR2 (**D**) and 6-OHDA ChR2 (**E**) mouse. (**F-I**) Effects of photoactivation on open field metrics for sham eYFP (n = 5), sham ChR2 (n = 6), 6-OHDA eYFP (n = 5), and 6-OHDA ChR2 (n = 5) groups (three-way MM ANOVAs, *post hoc* Bonferroni pairwise). *Photoactivation increased in both sham and 6-OHDA ChR2 groups:* (**F**) distance travelled (ChR2 vs. eYFP: $p < 0.01$), (**G**) locomotor bouts (ChR2 vs. eYFP: $p < 0.01$), (**H**) animal movement speed (ChR2 vs. eYFP: $p < 0.001$) and (**I**) duration of locomotion in the open field (ChR2 vs. eYFP: $p < 0.001$). (**J**) The graph presents animal rotational bias using the turn angle sum. There was a significant increase in 6-OHDA ChR2 rotational bias during A13 region photoactivation (6-OHDA ChR2 vs. 6-OHDA eYFP: $p < 0.001$). (**K**) Diagram depicting the pole test. A mouse is placed on a vertical pole facing upwards time for release is derived from time the experimenter removes their hand from the animal's tail to when it touches the ground. (**L, M**) Photoactivation of the A13 region led to shorter descent times in 6-OHDA ChR2 mice compared to 6-OHDA eYFP mice ($p < 0.01$). A pre-op baseline was performed, followed by post-op three weeks later. On experiment day, performance with no stim (Exp - NS) was compared with photostimulation (Exp - Stim). (**M**) 6-OHDA ChR2 mice showed a greater reduction in descent time compared to 6-OHDA eYFP mice (6-OHDA ChR2 vs. 6-OHDA eYFP: $p < 0.001$). ***$p < 0.001$, **$p < 0.01$. Error bars indicate SEMs.

the increased rotational bias was due to photoactivation or the interaction of photoactivation and the lesion. The sham ChR2 group showed no significant change in TAS with photoactivation (**Figure 3J**, $p = 0.06$). Next, we examined whether the increased turning angle sum in the 6-OHDA ChR2 group was observed during periods of locomotion. When the TAS was calculated only during periods of locomotion, the rotational bias in the animal orientation in the 6-OHDA ChR2 animals was not observed. The rotational bias was observed between bouts of locomotor activity.

Skilled vertical locomotion is improved in the pole test with photoactivation of the A13 region

The pole test is a classic 6-OHDA behavioral paradigm (**Figure 3K**) that involves skilled locomotor abilities for an animal to turn and descend a vertical pole (Matsuura et al., 1997; Ogawa et al., 1985). Improvements in function can be inferred if the time taken to complete the test decreases (Matsuura et al., 1997; Ogawa et al., 1985). 6-OHDA mice demonstrated significantly greater descent times than sham mice (**Figure 3L**, $p < 0.01$). Photoactivation of the A13 region reduced descent times for both 6-OHDA and sham groups on the pole test (**Figure 3L**, $p < 0.01$, Movie S3). Neither of the eYFP groups showed any changes in the time to complete the pole test.

To further understand the effects of photoactivation on the ability of mice to descend the pole, the time taken for mice to descend after turning was analyzed to remove any influence of animals spending time investigating their environment on the top of the pole. While all groups showed reduced total pole test descent time with photoactivation, considering just the time to descend from turn alone, there was a larger improvement with A13 region photoactivation in the 6-OHDA ChR2 mice compared to 6-OHDA eYFP mice (**Figure 3M**, $p < 0.001$). These results indicate that photoactivation has the effect of reducing bradykinesia by improving the ability of mice to descend the pole during the PT.

Dopaminergic Cells in the A13 region are preserved in the unilateral 6-OHDA mouse model

While photoactivation of the A13 region increased locomotor activity in both sham and 6-OHDA lesioned mice, we observed differences in speed and directional bias between the two groups. We hypothesized that these differences might be due to changes in the A13 region's connectome as previous research has shown that 6-OHDA lesions can lead to increases in firing frequency and metabolic activity in this brain region, over a period of one month (Périer et al., 2000). To investigate this possibility, we used whole brain imaging approaches (Hansen et al., 2020; Zhan et al., 2021) to examine changes in the connectome following 6-OHDA lesions of the nigrostriatal pathway.

As expected (Iancu et al., 2005), our whole brain imaging results showed that TH^+ cells in SNc (**Figure S4A-C**) were more vulnerable to the 6-OHDA neurotoxin than those in the ventral tegmental area (VTA) and A13 (**Figure S4D-F**). Specifically, we found that 6-OHDA-treated mice showed a significantly greater percentage of TH^+ cell loss in SNc compared to the VTA and A13 (VTA vs. SNc: $p < 0.01$; A13 vs. SNc: $p < 0.01$). In contrast, sham animals showed no significant difference in TH^+ cell loss across SNc, VTA and A13 (**Figure S4G**, $p > 0.05$). These findings are consistent with observations in the human brain, where the A13 region is preserved even in the presence of extensive nigrostriatal degeneration (Matzuk and Saper, 1985). Our results confirm that the 6-OHDA mouse model effectively replicates this aspect of Parkinson's disease pathology.

Extensive Remodeling of the A13 Region Connectome Following Unilateral Nigrostriatal Degeneration

Although photoactivation of the A13 region was effective in restoring speed in 6-OHDA lesioned mice (**Figure 3**), we observed an increase in circling behavior. This suggested that additional changes possibly reflecting alterations in the A13 connectome may be occuring. To investigate these potential changes without the potential confounds of an implanted optrode over the area, we conducted separate experiments to examine the changes in the input and output patterns of the A13 region. We did this by co-injecting anterograde (AAV8-CamKII-mCherry) and retrograde AAV (AAVrg-CAG-GFP) tracers into the A13 nucleus (Paxinos and Franklin, 2008) (**Figure S4H**). The injection core and spread were determined in the rostrocaudal direction from the injection site (**Figure S5**). The viral spread was centered around the zona incerta containing the A13, with minor spread to adjacent areas in some cases (**Figure S5**. To assess whether unilateral nigrostriatal degeneration led to changes in the organization of motor-related inputs and outputs from the A13, we first visualized interregional correlations of afferent and efferent proportions for each condition using correlation matrices (**Figure 4A and B**; 18 regions in a pairwise manner). Correlation matrices were organized using the hierarchical anatomical groups from the Allen Brain Atlas (**Figure 4C-F**). To control for experimental variations in the total labeling of neurons and fibers, we calculated the proportion of total inputs and outputs by dividing the afferent cell counts or efferent fiber areas in each brain region by the total number found in the brain. The data were then normalized to a $\log_{10}$ value to reduce variability and bring brain regions with high and low proportions of cells and fibers to a similar scale (Kimbrough et al., 2020). Comparing the afferent and efferent proportions pairwise between mice showed good consistency with an average correlation of 0.91 ± 0.02 (Spearman's correlation, **Figure S6**).

We observed changes in afferent input patterns to the A13 region between the sham and the 6-OHDA groups. Using a correlation matrix, we found that inputs from motor related brain regions to the A13 region in sham animals showed overall higher positive correlations across cortical, subcortical and brainstem areas than in 6-OHDA mice (**Figure 4C**). In 6-OHDA-lesioned mice, input distribution from dorsal pallidum, lateral hypothalamus, zona incerta, and tegmental reticular nucleus became positvely correlated to somatomotor (MOp, MOs) and somatosensory (SSp) cortical areas (**Figure 4D**). These regions were notably different (anti-correlated) compared to other brain regions. These data suggest a shift in the A13 inputs after 6-OHDA lesions, with a relative increase in cortical, pallidal, and hypothalamic influence compared to from other motor related brain regions.

In contrast, output patterns from the A13 region showed a higher overall correlation between brain regions in 6-OHDA-lesioned mice. In sham animals, A13 outputs to cortical regions were negatively correlated with outputs to thalamic, hypothalamic, and midbrain regions (**Figure 4E**). This pattern disappeared (**Figure 4F**) after nigrostriatal degeneration, suggesting a broader, less biased distribution of A13 outputs.

Differential remodeling of A13 region connectome ipsi- and contra-lesion following 6-OHDA-mediated nigrostriatal degeneration

We used whole-brain imaging to further investigate how the A13 connectome is affected by a unilateral 6-OHDA lesion. Our analysis revealed distinct patterns of remodelling on the lesioned (ipsilesional) and intact (contralesional) sides of the brain (**Figure 5B-E**). On the ipsilesional side, we observed a decrease in A13 afferent density in 6-OHDA mice compared to sham animals from several key regions (**Figure 5A**), including the primary motor cortex (MOp), primary somatosensory area (SSp), and secondary motor cortex (MOs). In contrast, ipsilesional compensatory increases in A13 afferents were observed, for example, in lateral hypothalamic area (LHA), Substantia Nigra pars reticulata (SNr), Superior colliclus (SCm), and PAG.

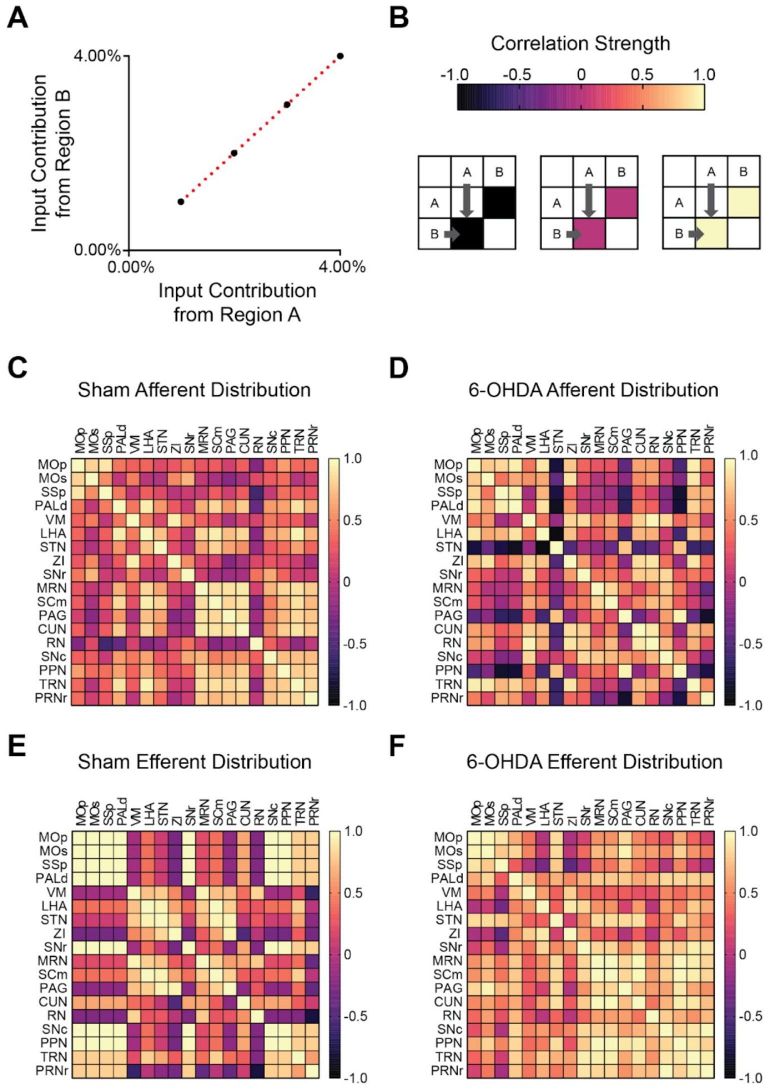


Figure 4.

Nigrostriatal degeneration causes widespread changes in A13 region input and output connections.

Correlation matrices were used to visualize input and output patterns of the A13 region. Here, we focus on motor-related pathways. **(A)** Brain regions with similar input patterns show high correlation. **(B)** Correlation strength is shown by cell color in the matrix: yellow represents strong positive correlations, magenta indicates no correlation, and black represents strong negative correlations. (C, D) Sham animals displayed stronger interregional correlations of inputs from motor-related regions across the neuroaxis to the A13 region compared to 6-OHDA-lesioned mice. This suggests a broader distribution of inputs among motor related cortical, subcortical, and brainstem regions in sham condition. (D) In 6-OHDA-lesioned mice, inputs to the A13 region from the STN, PAG, and PPN became negatively correlated, unlike inputs from other motor-related regions. In contrast, inputs from motor-related pallidal and incertohypothalamic areas showed stronger positive correlations with cortical inputs, suggesting these regions had a relatively greater influence on A13 activity. (E, F) In contrast, output patterns from the A13 region showed strong interregional correlation among cortical and brainstem motor related regions in 6-OHDA-lesioned mice compared to sham animals. (E) In sham animals, A13 outputs to cortical regions were negatively correlated with outputs to thalamic, hypothalamic, and midbrain regions. This pattern disappeared after nigrostriatal degeneration, suggesting a more widespread distribution of A13 outputs. MOp (primary motor cortex), MOs (secondary motor cortex), SSp (primary somatosensory area), PALd (pallidum, dorsal), VM (ventral medial thalamic nucleus), LHA (lateral hypothalamus), STN (subthalamic nucleus), ZI (zona incerta), SNr (substantia nigra pars reticular), MRN (midbrain reticular nucleus), SCm (superior colliculus, motor), PAG (periaqueductal gray), CUN (cuneiform), RN (red nucleus), SNc (substantia nigra pars compacta), PPN (pedunculopontine nucleus), TRN (tegmental reticular nucleus), and PRNr (pontine reticular nucleus).

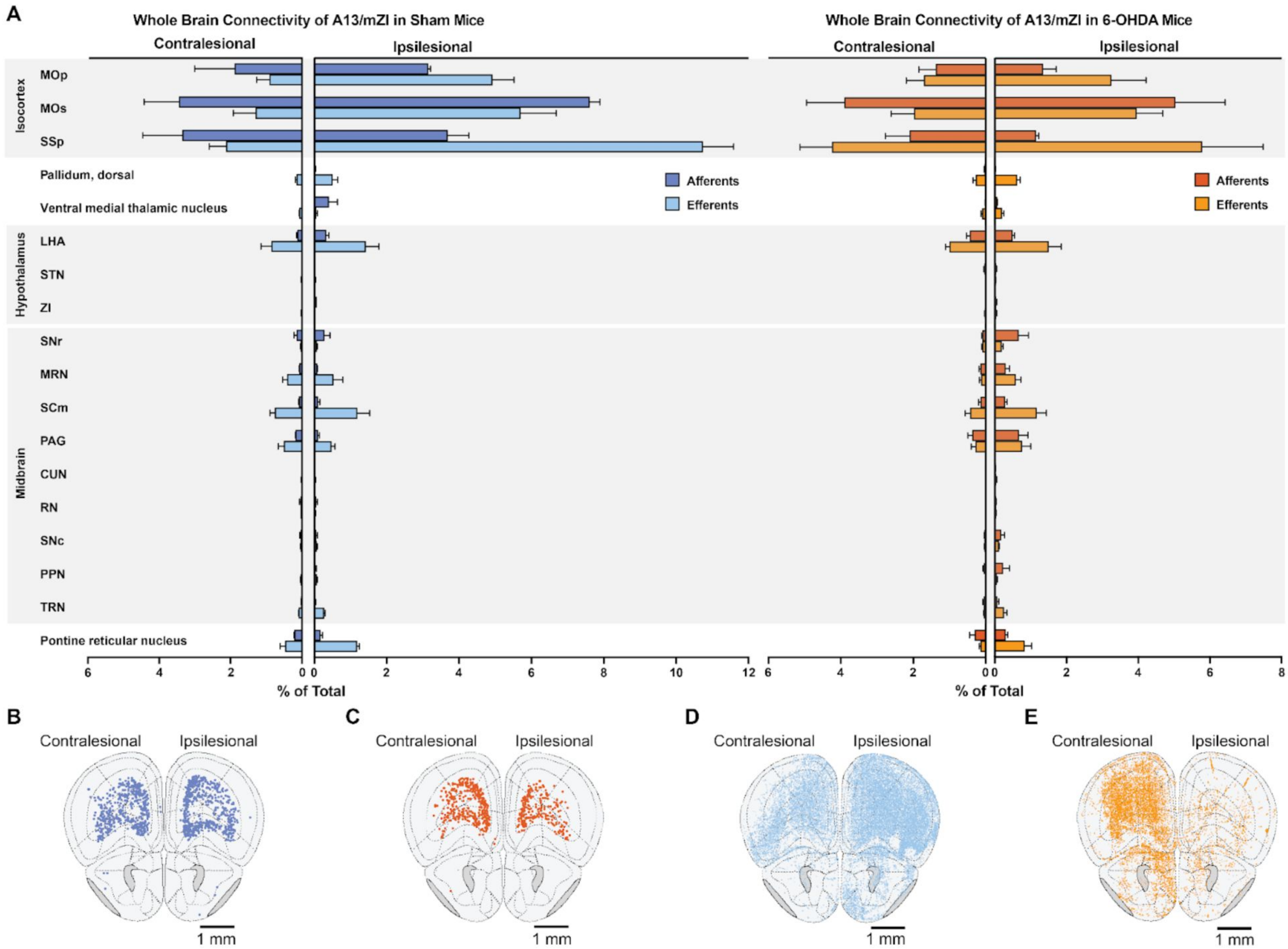


Figure 5.

Unilateral nigrostriatal degeneration causes distinct changes in A13 connectivity.

(A) A13 input and output connections to motor-related brain areas. Anterograde and retrograde viruses were injected into the A13 region on the lesioned side (see methods). The graph shows the relative change in input (afferents) and output (efferents) connections in 6-OHDA-lesioned mice compared to sham controls. **(B, C)** Brain regions showing changes in A13 input connections in sham **(B)** and 6-OHDA-lesioned **(C)** mice**. (D, E)** Brain regions showing changes in A13 output connections in sham **(D)** and 6-OHDA-lesioned **(E)** mice. Error bars represent SEMs. 6-OHDA: n = 3; sham: n = 2. Brain region abbreviations are from the Allen Brain Atlas. MOp (primary motor cortex), MOs (secondary motor cortex), SSp (primary somatosensory area), LHA (lateral hypothalamus), STN (subthalamic nucleus), ZI (zona incerta), SNr (substantia nigra pars reticulata), MRN (midbrain reticular nucleus), SCm (superior colliculus, motor), PAG (periaqueductal gray), CUN (cuneiform), RN (red nucleus), SNc (substantia nigra pars compacta), PPN (pedunculopontine nucleus), and TRN (tegmental reticular nucleus).

Interestingly, the contralesional side showed a more conservative pattern of afferent remodeling, with a modest decrease in A13 afferent density in primary motor cortex (MOp) and primary somatosensory cortex (SSp) of 6-OHDA mice, with a slight increase in the secondary motor cortex (MOs). For hypothalamic and midbrain structures only LHA and PAG showed evidence for an increase. Several regions, including the hypothalamus, midbrain, and pons showed bilateral upregulation of A13 afferents suggesting a more global compensatory response to the unilateral lesion.

Ipsilesional A13 efferents were decreased in 6-OHDA mice (**Figure 5A**) mainly in the somatosensory cortex (SSp), with some modest increases in midbrain structures, whereas, contralesional efferent projection patterns showed increases in efferent density 6-OHDA mice compared to sham in cortical structures (MOp, MOs, SSp). This was accompanied by modest decreases in efferent density in 6-OHDA contralesional midbrain structures compared to sham.

Discussion

Photoactivating the A13 region enhances locomotion in lesioned and sham mice, with increased distance travelled, locomotion duration, and speed. Particularly in 6-OHDA mice, it rescued the number of locomotor bouts and significantly improved bradykinesia. Additionally, we observed remodeling of the A13 connectivity post-nigrostriatal lesions. The zona incerta is a thin structure that extends from the rostral thalamus to the rostral portion of the red nucleus in rodents, and zona incerta regions around the subthalamic nucleus, situated substantially caudally to the A13, have shown clinical significance (Ossowska, 2019).

The role of the A13 region in locomotion in sham mice

We provide the first direct evidence that photoactivation of the A13 region can drive locomotion, suggesting the pro-locomotor functions of the zona incerta extend further rostrally in the mouse (Mitrofanis, 2005): photoactivation of cZI neurons increased animal movement speed in prey capture (Zhao et al., 2019) and active avoidance (Hormigo et al., 2020). Interestingly, early research on the cat suggested a role for the zona incerta in locomotion, particularly in the region adjacent to the subthalamic nucleus, but the effectiveness of rostral zona incerta regions containing the A13 was not reported (Grossman, 1958). Recent work shows that the A13 is involved in forelimb grasping (Garau et al., 2023), suggesting other motor control functions, while a positive valence associated with motivated food seeking behavior has also been reported (Ye et al., 2023). Previous work targeting the mZI region, including somatostatin (SOM^+), calretinin (CR^+), and $vGlut2^+$ neurons, did not change locomotor distance travelled in the OFT (Z. Li et al., 2021). However, multiple populations being photostimulated or targeting more medial populations in the ZI may contribute to the differences. Our findings align with mini-endoscope recordings from $CaMKII\alpha^+$ rostral ZI cells, which overlap the A13 showing subpopulations whose activity correlates with either movement speed or anxiety-related locations (Z. Li et al., 2021). Other work has found that photoactivation of GABAergic mZI pathways which project to the cuneiform promote exploratory activity by inhibiting cuneiform vGlut2 neurons (Sharma et al., 2024). The prolocomotory patterns following activation of mZI GABA populations and the more general activation of the region found in this work differ. The activity patterns with CamII kinase promotor transfection of the region produced a marked effect on locomotor speed, accompanied by thigmotactic behaviors not observed when mZI GABAergic populations are activated. Prolocomotory observed from the mZI region by us and others differ from the effects seen when lateral GABAergic ZI populations (dorsal and ventral ZI) are stimulated. Microinjection of $GABA_A$ receptor agonists, into the ZI results in a significant reduction in locomotor distance and velocity or cataplexy (Chen et al., 2023; Wardas et al., 1988). Alternatively, suppression of GABAergic ZI activity can increase locomotion following microinjection of $GABA_A$ receptor antagonists (Périer et al., 2002) or induce bradykinesia and akinesia by chemogenetic or optogenetic inhibition in healthy naive mice (Chen et al., 2023). The differing results likely are due to the projection patterns of GABAergic populations.

Because the A13 region photoactivation produces increased locomotor speed and improved descent times on the pole test, it highlights its role in movement control. Since A13 stimulation did not alter coordination during the task, it suggests a complex behavioral role consistent with its upstream location from the brainstem and its extensive connectome. A13 photoactivation increases animal speed, duration, and distance travelled. Taken together, these findings represent a rescue of function in the 6-OHDA model. Interestingly, in 6-OHDA and sham mice there was a latency of between 10 and 15 s on average following photostimulation before locomotion was initiated. Such delays are typical of stimulating sites upstream of the cuneiform, such as the dlPAG, which has a delay of several seconds (Tsang et al., 2021).

Photoactivation of the A13 reduces bradykinesia and akinesia in mouse PD models

While much work has targeted basal ganglia structures to address PD symptoms (DeLong and Wichmann, 2015), our research demonstrates that photoactivation of the A13 region can alleviate both bradykinesia and akinesia in 6-OHDA mice. Our work shows that A13 projections are affected at cortical and striatal levels following 6-OHDA, consistent with our observed changes in locomotor function. Over 28 days, there was a remarkable change in the afferent and efferent A13 regional connectome, despite the preservation of TH^{+} ZI cells. This is consistent with previous reports of widespread connectivity of the ZI (Mitrofanis, 2005). The preservation of A13 is expected since A13 lacks DAT expression (Bolton et al., 2015; Negishi et al., 2020; Sharma et al., 2018) and is spared from DAT-mediated toxicity of 6-OHDA (Dauer and Przedborski, 2003; Konnova et al., 2018; Simola et al., 2007). While A13 cells were spared following nigrostriatal degeneration, our work demonstrates its connectome was rewired. The ipsilateral afferent projections were markedly downregulated, while contralesional projecting afferents showed upregulation. In contrast, efferent projections showed less downregulation in the cortical subplate regions and bilateral upregulation of thalamic and hypothalamic efferents. Similar timeframes for anatomical and functional plasticity affecting neurons and astrocytes following an SNc or MFB 6-OHDA have been previously reported (Bosson et al., 2015; Perović et al., 2005; Requejo et al., 2020). Human PD brains that show degeneration of the SNc have a preserved A13 region, suggesting that our model, from this perspective, is externally valid (Matzuk and Saper, 1985).

Combined with photoactivation of the A13 region, we provide evidence for plasticity following damage to SNc. A previous brain-wide quantification of TH levels in the MPTP mouse model identified additional complexity in regulating central TH expression compared to conventional histological studies (Roostalu et al., 2019). These authors reported decreased SNc TH^{+} cell numbers without a significant change in TH^{+} intensity in SNc and increased TH^{+} intensity in limbic regions such as the amygdala and hypothalamus (Roostalu et al., 2019). Still, there was a downstream shift in the distribution pattern of A13 efferents following nigrostriatal degeneration with a reduction in outputs to cortical and striatal subregions. This suggests A13 efferents are more distributed across the neuraxis than in sham mice. One hypothesis arising from our work is that the preserved A13 efferents could provide compensatory innervation with collateralization mediated contralesionally and, in some subregions, ipsilesionally to increase the availability of extracellular dopamine.

Several A13 efferent targets could be responsible for rotational asymmetry. In a unilateral 6-OHDA model, ipsiversive circling behaviour is indicative of intact striatal function on the contralesional side (Carey, 1991; Schwarting et al., 1991; Ungerstedt, 1971; Zetterström et al., 1986). Instead, the predictive value of a treatment is determined by contraversive circling mediated by increased dopamine receptor sensitivity on the ipsilesional striatal terminals (Costall et al., 1976; Lane et al., 2006). Our findings show that A13 stimulation enhances ipsiversive circling and may represent a gain of function on the intact side, but this may be simply due to 6-OHDA mice having reduced locomotion overall. Given the preservation of A13 cells in PD, bilateral stimulation of A13 could potentially reduce motor asymmetry and alleviate bradykinesia and akinesia.

With the induction of a 6-OHDA lesion, there is a change in the A13 connectome, characterized by a reduction in bidirectional connectivity with ipsilesional cortical regions. In rodent models, the motor cortices, including the M1 and M2 regions, can shape rotational asymmetry (Gradinaru et al., 2009; Magno et al., 2019; Sanders and Jaeger, 2016; Valverde et al., 2020). Activation of M1 glutamatergic neurons increases the rotational bias (Valverde et al., 2020), while M2 neuronal stimulation promotes contraversive rotations (Magno et al., 2019). Our data suggest that A13 photoactivation may have resulted in the inhibition of glutamatergic neurons in the contralesional M1. An alternative possibility is the activation of the contralesional M2 glutamatergic neurons, which would be expected to induce increased ipsilesional rotations (Magno et al., 2019). The ZI could generate rotational bias by A13 modulation of cZI glutamatergic neurons via incerto-incertal fibers (Ossowska, 2019; Power and Mitrofanis, 1999), which promotes asymmetries by activating the SNr (L.-X. Li et al., 2021). The incerto-incertal interconnectivity has not been well studied, but the ZI has a large degree of interconnectivity (Sharma et al., 2018; Tsang et al., 2021) along all axes and between hemispheres (Power and Mitrofanis, 1999). However, this may only contribute minimally given that unilateral photoactivation of the A13 cells in sham mice failed to produce ipsiversive turning behavior while unilateral photoactivation of cZI glutamatergic neurons in sham animals was sufficient in generating ipsiversive turning behavior (L.-X. Li et al., 2021). Another possibility involves the A13 region projections to the MLR. With the unknown downstream effects of A13 photoactivation, there may be modulation of the PPN neurons responsible for this turning behavior (Masini and Kiehn, 2022). The thigmotactic behaviors suggest some effects may be mediated through dlPAG and CUN (Tsang et al., 2021), and recent work suggests the CUN as a possible therapeutic target (Fougère et al., 2021; Noga and Whelan, 2022). Since PD is a heterogeneous disease, our data provide another therapeutic target providing context-dependent relief from symptoms. This is important since PD severity, symptoms, and progression are patient-specific.

Towards a preclinical model

To facilitate future translational work applying DBS to this region, we targeted the A13 region using AAV8-CamKII-mCherry viruses. The CaMKIIα promoter virus is beneficial because it is biased towards excitatory cells (Haery et al., 2019), narrowing the diversity of the transfected A13 region and when combined with traditional therapies, such as L-DOPA, it could be a translatable strategy (Watakabe et al., 2015; Watanabe et al., 2020). Optogenetic strategies have been used to activate retinal cells in humans, partially restoring visual function and providing optimism that AAV-based viral strategies can be adapted in other human brain regions (Sahel et al., 2021). A more likely possibility for stimulation of deep nuclei is that DREADD technology could be adapted, which would not require any implants, however, this remains speculative. Near-term, our work suggests the A13 is a possible target for DBS. Gait dysfunction in PD is particularly difficult to treat, and indeed when DBS of subthalamic nucleus is deployed, a mixture of unilateral and bilateral approaches have been used (Lizarraga et al., 2016), along with stimulation of multiple targets (Stefani et al., 2007). This represents the heterogeneity of PD and underlines the need for considering multiple targets. DBS does not always have the same outcomes as optogenetic stimulation (Neumann et al., 2023). Alternative stimulation strategies such as cyclic burst DBS (Fischer et al., 2020) may provide ways to augment or circumvent the need for optogenetic approaches to be deployed. Future work may want to consider a multipronged strategy to hone burst stimulation parameters with identification of cell populations to deploy DBS in a more targeted manner (Spix et al., 2021).

Limitations

Currently, few PD animal models are available that adequately model the progression and the extent of SNc cellular degeneration while meeting the face validity of motor deficits (Dauer and Przedborski, 2003; Konnova et al., 2018). While the 6-OHDA models fail to capture the age-dependent chronic degeneration observed in PD, they lead to robust motor deficits with acute degeneration and allow for compensatory changes in connectivity to be examined. Moreover, the 6-OHDA lesions resemble the unilateral onset (Hughes et al., 1992) and persistent asymmetry (Lee et al., 1995) of motor dysfunction in PD. Another option could be the MPTP mouse model, which offers the ease of systemic administration and translational value to primate models; however, the motor deficits are variable and lack the asymmetry observed in human patients (Hughes et al., 1992; Jagmag et al., 2015; Lee et al., 1995; Meredith and Rademacher, 2011). Despite these limitations, the neurotoxin-based mouse models, such as MPTP and 6-OHDA, offer greater SNc cell loss than genetic-based models; in the case of the 6-OHDA model, it captures many aspects of motor dysfunctions in PD (Dauer and Przedborski, 2003; Jagmag et al., 2015; Konnova et al., 2018; Simola et al., 2007). As in human PD, we found no significant change in A13 TH^{+} cell counts (Matzuk and Saper, 1985). Our work suggests a possibility that glutamatergic neurons may be involved, however future work will be required to tease apart the relative contributions of neuronal populations within the A13 region contributing to the effects shown here. A more comprehensive approach with opto-tagging specific cell classes and specific labeling of glutamatergic versus dopaminergic classes could be an option, especially considering work showing the presence of multiple classes of dopamine cells in the hypothalamus (Romanov et al., 2017) . Another limitation is that since A13 neurons remained intact following a lesion, it is possible that changes in connectome reflected secondary effects from other regions impacted by the 6-OHDA lesion. However, the fact that there was a significant change in the connectome post-6-OHDA injection and striatonigral degeneration is in and of itself important to document.

Conclusions

Our research underscores the role of the A13 region beyond the classic nigrostriatal axis in Parkinson’s disease, driving locomotor activity and mitigating bradykinetic and akinetic deficits linked to impaired DAergic transmission. This observation indicates a rescue of locomotion loss in 6-OHDA-lesioned mice, as well as bradykinesia. Additionally, it produced possible gain-of-function effects, such as circling behavior, which may be attributed to plasticity changes induced by the 6-OHDA lesions. Widespread remodelling of the A13 region connectome is critical to our understanding of the effects of dopamine loss in PD models. In summary, our findings support an exciting role for the A13 region in locomotion with demonstrated benefits in a mouse PD model and contribute to our understanding of heterogeneity in PD.

Materials and methods

Animals

All care and experimental procedures were approved by the University of Calgary Health Sciences Animal Care Committee (Protocol #AC19-0035). C57BL/6 male mice 49 - 56 days old (weight: M = 31.7 g, SEM = 2.0 g) were group-housed (< five per cage) on a 12-h light/dark cycle (07:00 lights on - 19:00 lights off) with *ad libitum* access to food and water, as well as cat’s milk (Whiskas, Mars Canada Inc., Bolton, ON, Canada). Mice were randomly assigned to the groups described.

Surgical Procedures

We established a well-validated unilateral 6-OHDA mediated Parkinsonian mouse model (Thiele et al., 2012) (**Figure 1**, Movie S4). 30 minutes before stereotaxic microinjections, mice were intraperitoneally injected with desipramine hydrochloride (2.5 mg/ml, Sigma-Aldrich) and pargyline hydrochloride (0.5 mg/ml, Sigma-Aldrich) at 10 ml/kg (0.9% sterile saline, pH 7.4) to enhance selectivity and efficacy of 6-OHDA induced lesions (Thiele et al., 2012). All surgical procedures were performed using aseptic techniques, and mice were anesthetized using isoflurane (1 - 2%) delivered by 0.4 L/min of medical-grade oxygen (Vitalair 1072, 100% oxygen).

Mice were stabilized on a stereotaxic apparatus. Small craniotomies were made above the medial forebrain bundle (MFB) and the A13 nucleus within one randomly assigned hemisphere. Stereotaxic microinjections were performed using a glass capillary (Drummond Scientific, PA, USA; Puller Narishige, diameter 15 – 20 mm) and a Nanoject II apparatus (Drummond Scientific, PA, USA). 240 nL of 6-OHDA (3.6 µg, 15.0 mg/mL; Tocris, USA) was microinjected into the MFB (AP -1.2 mm from bregma; ML ±1.1 mm; DV -5.0 mm from the dura). Sham mice received a vehicle solution (240 nL of 0.2% ascorbic acid in 0.9% saline; Tocris, USA).

Whole Brain Experiments

For tracing purposes, a 50:50 mix of AAV8-CamKII-mCherry (Neurophotonics, Laval University, Quebec City, Canada, Lot #820, titre 2×10^{13} GC/ml) and AAVrg-CAG-GFP (Addgene, Watertown, MA, Catalogue #37825, Lot #V9234, titre ≥ 7×10^{12} vg/mL) was injected ipsilateral to 6-OHDA injections at the A13 nucleus in all mice (AP -1.22 mm from bregma; ML ±0.4 mm; DV -4.5 mm from the dura, the total volume of 110 nL at a rate of 23 nL/s). Post-surgery care was the same for both sham and 6-OHDA injected mice. The animals were sacrificed 29 days after surgery.

Photoactivation Experiments

36.8 nL of AAVDJ-CaMKIIα-hChR2(H134R)-eYFP (UNC Stanford Viral Gene Core; Stanford, CA, US, Catalogue #AAV36; Lots #3081 and #6878, titres 1.9×10^{13} and 1.7×10^{13} GC/mL, respectively) or eYFP control virus (AAVDJ-CaMKIIα-eYFP; Lots #2958 and #5510, titres 7.64×10^{13} and 2.88×10^{13} GC/mL, respectively) were injected into the A13 (AP: -1.22 mm; ML -0.5 mm from the Bregma; DV -4.5 mm from the dura). A mono-fiber cannula (Doric Lenses, Quebec, Canada, Catalogue #B280-2401-5, MFC_200/230-0.48_5mm_MF2.5_FLT) was implanted slowly 300 µm above the viral injection site. Metabond® Quick Adhesive Cement System (C&B, Parkell, Brentwood, NY, US) and Dentsply Repair Material (Dentsply International Inc., York, PA, USA) were used to fix the optical fiber in place. Animals recovered from the viral surgery for 19 days before follow-up behavioral testing. **Figure 1** shows a timeline of behavioral tests.

ChR2 photoactivation

A Laserglow Technologies 473 nm laser and driver (LRS-0473-GFM-00100-05, North York, ON, Canada) were used to generate the photoactivation for experiments. The laser was triggered by TTL pulses from an A.M.P.I. Master-8 stimulator (Jerusalem, Israel) or an Open Ephys PulsePal (Sanworks, Rochester, NY, US) set to 20 Hz, 10-ms pulse width and 5 mW power. All fiber optic implants were tested for laser power before implantation (Thorlabs, Saint-Laurent, QC, Canada; optical power sensor (S130C) and meter (PM100D)). The Stanford Optogenetics irradiance calculator was used to estimate the laser power for stimulation ("Stanford Optogenetics Resource Center," n.d.). A 1×2 fiber-optic rotary joint (Doric Lenses, Quebec, Canada; FRJ_1×2i_FC-2FC_0.22) was used. The animals' behaviors were recorded with an overhead camera (SuperCircuits, Austin, TX, US; FRJ_1×2i_FC-2FC_0.22; 720 x 480 resolution; 30 fps). The video was processed online (Cleversys, Reston, VA; TopScan V3.0) with a TTL signal output from a National Instruments 24-line digital I/O box (NI, Austin, TX, US; USB-6501) to the Master-8 stimulator.

Behavioral Testing

Open Field Test

Each mouse was placed in a square arena measuring 70 (W) x 70 (L) x 50 (H) cm with opaque walls and recorded for 30 minutes using a vertically mounted video camera (Model PC165DNR, Supercircuits, Austin, TX, USA; 30 fps). 19 days following surgery, mice were habituated to the open field test (OFT) arena with a patch cable attached for three days in 30 minute sessions to bring animals to a common baseline of activity. On experimental days, after animals were placed in the OFT, a one-minute-on-three-minutes-off paradigm was repeated five times following an initial ten minutes baseline activity. Locomotion was registered when mice traveled a minimum distance of 10 cm at 6 cm/s for 20 frames over a 30-frame segment. When the mouse velocity dropped below 6 cm/s for 20 frames, locomotion was recorded as ending. Bouts of locomotion relate to the number of episodes where the animal met these criteria. Velocity data were obtained from the frame-by-frame results and further processed in a custom Python script to detect instantaneous speeds greater than 2 cm/s (Masini and Kiehn, 2022). All animals that had validated targeting of the A13 region were included in the OFT data presented in the results section, except for one sham ChR2 animal, which showed grooming rather than the typical locomotor phenotypes.

Pole Test

Mice were placed on a vertical wooden pole (50 cm tall and 1 cm diameter) facing upwards and then allowed to descend the pole into their home cage (Glajch et al., 2012). Animals were trained for three days and tested 2-5 days pre-surgery. Animals were acclimatized 21-22 days post-surgery under two conditions: without a patch cable and with the patch cable attached without photoactivation. On days 24-27, experimental trials were recorded with photoactivation. Video data were recorded for a minimum of three trials (Canon, Brampton, ON, Canada; Vixia HF R52; 1920 x 1080 resolution; 60 fps). A blinded scorer recorded the times for the following events: the hand release of the animal's tail, the animal fully turning to descend the pole, and the animal reaching the base of the apparatus. Additionally, partial falls, where the animal slipped down the pole but did not reach the base, and full falls, where the animal fell to the base, were recorded separately. All validated animals were included in the quantified data, including the sham ChR2 animal that began grooming in the OFT upon photoactivation. This animal displayed proficiency in performing the PT during photoactivation. It started grooming upon completion of the task when photoactivation was on. One sham ChR2 animal was photostimulated at 1 mW since it would jump off the apparatus at higher stimulation intensities.

Immunohistochemistry

A13 and SNc region

Post hoc analysis of the tissue was performed to confirm the 6-OHDA lesion and validate the targeting of the A13 region. Following behavioral testing, animals underwent a photoactivation protocol to activate neurons below the fiber optic tip (Koblinger et al., 2018). Animals were placed in an OFT for ten minutes before receiving three minutes of photoactivation. Ten minutes later, the animals were returned to their home cage. 90 minutes post photoactivation, animals were deeply anesthetized with isoflurane and then transcardially perfused with room temperature PBS followed by cold 4% paraformaldehyde (PFA) (Sigma-Aldrich, Catalogue #441244-1KG). The animals were decapitated, and the whole heads were incubated overnight in 4% PFA at 4°C before the fiber optic was removed and the brain removed from the skull. The brain tissue was post-fixed for another 6 - 12 hours in 4% PFA at 4°C then transferred to 30% sucrose solution for 48 - 72 hours. The tissue was embedded in VWR® Clear Frozen Section Compound (VWR International LLC, Radnor, PA, US) and sectioned coronally at 40 or 50 µm using a Leica cryostat set to -21°C (CM 1850 UV, Concord, ON, Canada). Sections from the A13 region (−0.2 to -2.0 mm past bregma) and the SNc (−2.2 to -4.0 mm past bregma) were collected and stored in PBS containing 0.02% (w/v) sodium azide (EM Science, Catalogue #SX0299-1, Cherry Hill, NJ, US) (Paxinos and Franklin, 2008).

Immunohistochemistry staining was done on free-floating sections. The A13 sections were labelled for c-Fos, TH, and GFP (to enhance eYFP viral signal), and received a DAPI stain to identify nuclei. The SNc sections were stained with TH and DAPI (**Table 1**). Sections were washed in PBS (3 x 10 mins) then incubated in a blocking solution comprised of PBS containing 0.5% Triton X-100 (Sigma-Aldrich, Catalogue #X100-500ML, St. Louis, MO, US) and 5% donkey serum (EMD Millipore, Catalogue #S30-100ML, Billerica, MA, USA) for 1 hour. This was followed by overnight (for SNc sections) or 24-hour (for A13 sections) incubation in a 5% donkey serum PBS primary solution at room temperature. On day 2, the tissue was washed in PBS (3 x 10 mins) before being incubated in a PBS secondary solution containing 5% donkey serum for 2 hours (for SNc tissue) or 4 hours (for A13 tissue). The secondary was washed with a PBS solution containing 1:1000 DAPI for 10 mins, followed by a final set of PBS washes (3 x 10 mins). Tissue was mounted on Superfrost® micro slides (VWR, slides, Radnor, PA, US) with mounting media (Vectashield®, Vector Laboratories Inc., Burlingame, CA, US), covered with #1 coverslips (VWR, Radnor, PA, US) then sealed.

Whole Brain

Mice were deeply anesthetized with isoflurane and transcardially perfused with PBS, followed by 4% PFA. To prepare for whole brain imaging, brains were first extracted and postfixed overnight in 4% PFA at 4°C. The next day, a modified iDISCO method (Renier et al., 2014) was used to clear the samples and perform quadruple immunohistochemistry in whole brains. The modifications include prolonged incubation and the addition of SDS for optimal labelling. Antibodies used are listed in **Table 1** and the protocol is provided in **Table 2**.

Image Acquisition and Analysis

Photoactivation Experiments

All tissue was initially scanned with an Olympus VS120-L100 Virtual Slide Microscope (UPlanSApo, 10x and 20x, NA = 0.4 and 0.75). Standard excitation and emission filter cube sets were used (DAPI, FITC, TRITC, Cy5), and images were acquired using an Orca Flash 4.0 sCMOS monochrome camera (Hamamatsu, Bridgewater Township, NJ, US). For c-Fos immunofluorescence, A13 sections of the tissue were imaged with a Leica SP8 FALCON (FAst Lifetime CONtrast) scanning confocal microscope equipped with a tunable laser and using a 63x objective (HC PlanApo, NA = 1.40).

SNc images were imported into Adobe Illustrator, where the SNc (Fougère et al., 2021), including the pars lateralis (SNl), was delineated using the TH immunostaining together with the medial lemniscus and cerebral peduncle as landmarks (bregma -3.09 and -3.68 mm) (Iancu et al., 2005; Paxinos and Franklin, 2008; Stott and Barker, 2014). Cell counts were obtained using a semi-automated approach using an Ilastik (v1.4.0b15) (Berg et al., 2019) trained model followed by corrections by a blinded counter (Fougère et al., 2021; Iancu et al., 2005). Targeting was confirmed on the 10x overview scans of the A13 region tissue by the presence of eYFP localized in the mZI around the A13 TH^+ nucleus, the fiber optic tip being visible near the mZI and A13 nucleus, and the presence of c-Fos positive cells in $ChR2^+$ tissue. C-Fos expression colocalization within the A13 region was performed using confocal images. The mZI & A13 region was identified with the 3rd ventricle and TH expression as markers (Paxinos and Franklin, 2008).

Primary Antibody	Species	Type	Supplier	Catalogue #	Dilution	Tissue Region
c-Fos	Rabbit	Polyclonal	Synaptic Systems (Göttingen, Germany)	226 003	1:1000	A13
GFP	Chicken	Polyclonal	Aves Lab (Tigard, OR)	GFP-1010	1:1000 1:5000	Whole brain A13
mCherry	Rat	Monoclonal	Invitrogen, Thermo Fisher Scientific (Waltham, MA)	M11217	1:500	Whole brain
TH	Rabbit	Polyclonal	Abcam (Cambridge, UK)	AB112	1:500 1:1000	Whole brain SNc
TH	Sheep	Polyclonal	Millipore Sigma (Burlington, MA)	AB1542	1:500	A13
Secondary Antibody			**Supplier**	**Catalogue #**	**Dilution**	**Tissue Region**
Alexa Fluor® 488 Donkey Anti-Chicken			Jackson ImmunoResearch Inc. (Westgrove, PA)	703-545-155	1:200 1:1000	Whole brain A13
Alexa Fluor® 594 Donkey Anti-Rabbit			Invitrogen, Thermo Fisher Scientific	A-21207	1:500	A13
Alexa Fluor® 647 Donkey Anti-Rabbit			Invitrogen, Thermo Fisher Scientific	A-31573	1:1000	SNc
Alexa Fluor® 647 Donkey Anti-Sheep			Invitrogen, Thermo Fisher Scientific	A-21448	1:1000	A13
Alexa Fluor® 790 Donkey Anti-Rabbit			Invitrogen, Thermo Fisher Scientific	A-11374	1:200	Whole brain
Cy™3 AffiniPure Donkey Anti-Rat			Jackson ImmunoResearch Inc.	712-165-153	1:200	Whole brain
Nuclear Stain			**Supplier**	**Catalogue #**	**Dilution**	**Tissue Region**
DAPI			Invitrogen, Thermo Fisher Scientific	D1306	1:1000	A13, SNc
TO-PRO™-3 Iodide (642/661)			Invitrogen, Thermo Fisher Scientific	T3605	1:5000	Whole brain

Table 1.

List of antibodies used for immunohistochemical staining of the A13 and SNc regions, as well as the whole brain.

Day #	Instructions
1	Washed 2 x 30 minutes in PBS on a shaker at room temperature.
2	Dehydrated tissue in methanol/H_2O series of 20%, 40%, 60%, 80%, 100%, 100% (1 hour each at room temperature) then left overnight in a 66% dichloromethane/33% methanol solution.
3	Washed twice in 100% methanol at room temperature and then chilled the samples at 4°C. Bleached the samples in chilled fresh 5% H_2O_2 in methanol (1 volume 30% H_2O_2 to 5 volumes methanol), overnight at 4°C.
4	Rehydrated in methanol/water series of 80%, 60%, 40%, 20%, PBS (1 hour each at room temperature). Washed twice in PTx.2 (0.2% Triton™ X-100 in PBS). Incubated the samples in Permeabilization Solution at 37°C for 2 days.
6	Washed 3 x 1-2 hours in 0.5mM SDS/PBS at 37°C then incubated for 3 days.
9	Incubated with the primary antibody in 0.5mM SDS/PBS at 37°C for 3 days.
12	Refreshed with the primary antibody in PTx.2 at 37°C then incubated for 4 days.
17	Washed 5 x 2 hours in PTwH (0.2% Tween® 20 and 10 mg/mL Heparin stock solution in PBS) and incubated at 37°C overnight.
18	Incubated with secondary antibody in PTwH/3% Donkey Serum at 37°C for 3 days.
21	Refreshed with secondary antibody in PTwH/3% Donkey Serum at 37°C for 4 days.
25	Washed 5 x 2 hours in PTwH then incubated at 37°C overnight.
26	Dehydrated in methanol/water series of 20%, 40%, 60%, 80%, 100% (1 hour each at room temperature). Refreshed the samples with 100 % methanol and left overnight at room temperature.
27	Incubated in 66% DCM/33% methanol for 3 hours on a shaker at room temperature. Then incubated in 100% DCM (Sigma 270997-12X100ML) for 2 x 15 minutes (with shaking) to wash away methanol. Incubated samples with ethyl cinnamate for 3 hours at room temperature with shaking. Refreshed ethyl cinnamate then left at room temperature for imaging.

Table 2.

Protocol for Whole Brain Clearing.

Whole Brain Experiments

Cleared whole brain samples were imaged using a light-sheet microscope (LaVision Biotech UltraMicroscope, LaVision, Bielefeld, Germany) with an Olympus MVPLAPO 2x objective with 4x optical zoom (NA = 0.475) and a 5.7 mm dipping cap that is adjusted for the high refractive index of 1.56. The brain samples were imaged in an ethyl cinnamate medium to match the refractive indices and illuminated by three sheets of light bilaterally. Each light sheet was 5 μm thick, and the width was set at 30% to ensure sufficient illumination at the centroid of the sample. Laser power intensities and chromatic aberration corrections used for each laser were as follows: 10% power for 488 nm laser, 5% power for 561 nm laser with 780 nm correction, 40% power for 640 nm laser with 960 nm correction, and 100% power for 785 nm laser with 1,620 nm correction. Each sample was imaged coronally in 8 by 6 squares with 20% overlap (10,202 μm by 5,492 μm in total) and a z-step size of 15 μm (xyz resolution = 0.813 μm x 0.813 μm x 15 μm). While an excellent choice for our work, confocal microscopy offers better resolution at the expense of time. To gain a better resolution using a light-sheet microscope in select regions (eg. SNc and A13 cells), we increased the optical zoom to 6.3x.

A13 Connectome Analysis

Images were processed using ImageJ software (Schneider et al., 2012). Raw images were stitched, and a z-encoded maximum intensity projection across a 90 μm thick optical section was obtained across each brain. 90 μm sections were chosen because the 2008 Allen reference atlas images are spaced out at around 100 μm. Brains with insufficient quality in labeling were excluded from analysis (*n* = 1 of three sham and *n* = 3 of six 6-OHDA mice). Instructions for identifying YFP^+ or TH^+ cells to annotate were provided to the manual counters. YFP^+ and TH^+ cells were manually counted using the Cell Counter Plug-In (ImageJ). $mCherry^+$ fibers were segmented semi-automatically using Ilastik software (Berg et al., 2019) and quantified using particle analysis in ImageJ. Images and segmentations were imported into WholeBrain software to be registered with the 2008 Allen reference atlas (Fürth et al., 2018). The TO-PRO™-3 and TH channels were used as reference channels to register each section to a corresponding atlas image. ImageJ quantifications of cell and fiber segmentations were exported in XML formats and registered using WholeBrain software. To minimize the influence of experimental variation on the total labeling of neurons and fibers, the afferent cell counts or efferent fiber areas in each brain region were column divided by the total number found in a brain to obtain the proportion of total inputs and outputs. Connectome analyses were performed using custom R scripts (L. H. Kim et al., 2021). For interregional correlation analyses, the data were normalized to a $\log_{10}$ value to reduce variability and bring brain regions with high and low proportions of cells and fibers to a similar scale. The consistency of afferent and efferent proportions between mice was compared in a pairwise manner using Spearman's correlation (**Figure S6**).

Quantification of 6-OHDA mediated TH^+ cell loss

The percentage of TH^+ cell loss was quantified to confirm 6-OHDA mediated SNc lesions. TH^+ cells within ZI, VTA and SNc areas from 90 μm thick optical brain slice images (AP: -0.655 to -3.88 mm from bregma) were manually counted by two blinded counters (*n* = 3 sham and *n* = 6 6-OHDA mice; ZI region in 2 of 6 6-OHDA mice were excluded due to presence of abnormal scarring/healing at the injection site of viruses). Subsequently, WholeBrain software (Fürth et al., 2018) was used to register and tabulate TH^+ cells in the contralesional and ipsilesional brain regions of interest. Counts obtained from the two counters were averaged per region. The percentage of TH^+ cell loss was calculated by dividing the difference in counts between contralesional and ipsilesional sides by the contralesional side count and multiplying by 100%.

Statistical analyses

All data were tested for normality using a Shapiro-Wilk test to determine the most appropriate statistical tests. The percent ipsilesional TH^+ neuron loss within the SNc as defined above using a Pearson correlation (Fougère et al., 2021) was used to ascertain the effect of the 6-OHDA lesion on behavior. A Wilcoxon rank-sum test was performed for comparisons within subjects at two timepoints where normality failed, and the central limit theorem could not be applied. The two groups were compared using an unpaired t-test with Welch's correction. A mixed model ANOVA (MM ANOVA) was used to compare the effects of group type, injection type and time. Additionally, Mauchly's test of sphericity was performed to account for differences in variability within the repeated measures design. A Greenhouse Geisser correction was applied to all ANOVAs where Mauchly's test was significant for RM and MM ANOVAs. The *post hoc* multiple comparisons were run when the respective ANOVAs reached significance using Dunnett's or Dunn's tests for repeated measures of parametric and non-parametric tests, respectively. The pre-stimulation timepoints were used as the control time point to determine if stimulation altered behavior. A Bonferroni correction was added for *post hoc* comparisons following a MM ANOVA between groups at given time points to control for alpha value inflation. All correlations, t-tests, and ANOVAs were performed, and graphs were created using Prism version 9.3.1 (Graphpad) or SPSS (IBM, 28.0.1.0). Full statistical reporting is in Supplemental Statistics.xls.

Figures were constructed using Adobe Photoshop and Illustrator.

Supplementary material

Figure S1.

Quantification of channelrhodopsin viral spread in the rostral-caudal direction from the injection site in 6-OHDA-treated and sham animals.

The left panel shows a representative image of viral spread, including the optic fibre track, visualized using the “fire” lookup table in FIJI/ImageJ software, with targeting precision confirmed by coronal overlays from the Mouse Allen Brain Atlas to the A13/Mzi region. The right panel displays a graph illustrating the percentage of viral spread across anterior-posterior coordinates for all ChR2-infected mice, calculated as the ratio of viral expression area to total tissue section area. Data were obtained from mouse brain sections injected with AAVDJ-CaMKIIα-ChR2 into the A13/Mzi region, sectioned at 50 μm, and analyzed using the VS120 Virtual Slide Scanner. Quantification was performed by blinded analysts using FIJI/ImageJ, with defined regions of interest (ROIs).

SUPPLEMENTARY MATERIAL

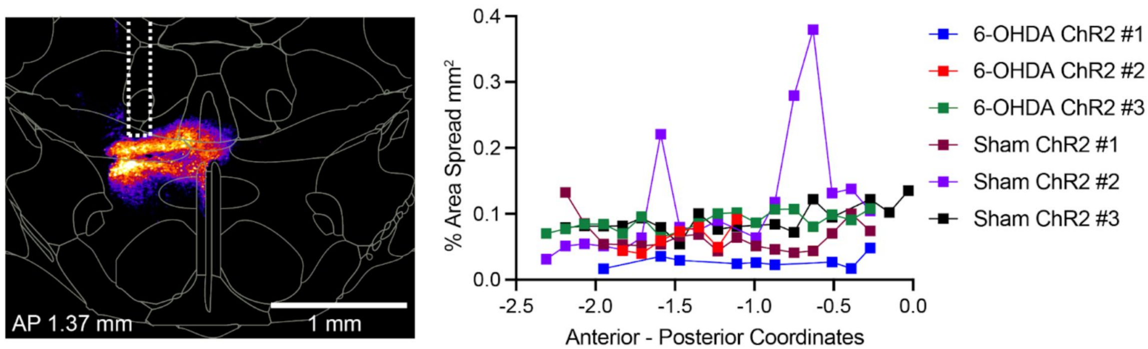
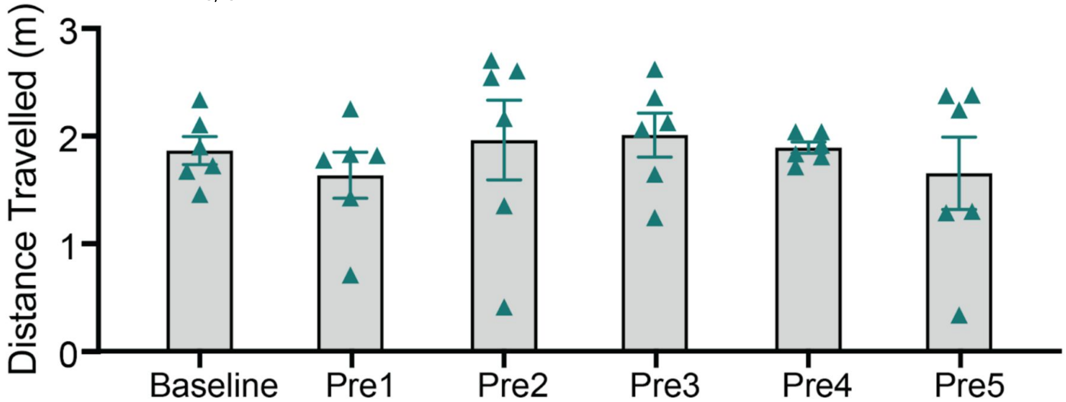


Figure S2.

Time course of open field locomotion distance travelled over 30 minutes.

Locomotion distance travelled for the six sham ChR2 animals at baseline and at the five pre-stimulation timepoints compared using a 1-way RM ANOVA ($F_{5,25}$ = 0.49, p = 0.78). Data indicate mean ± SEM bars.



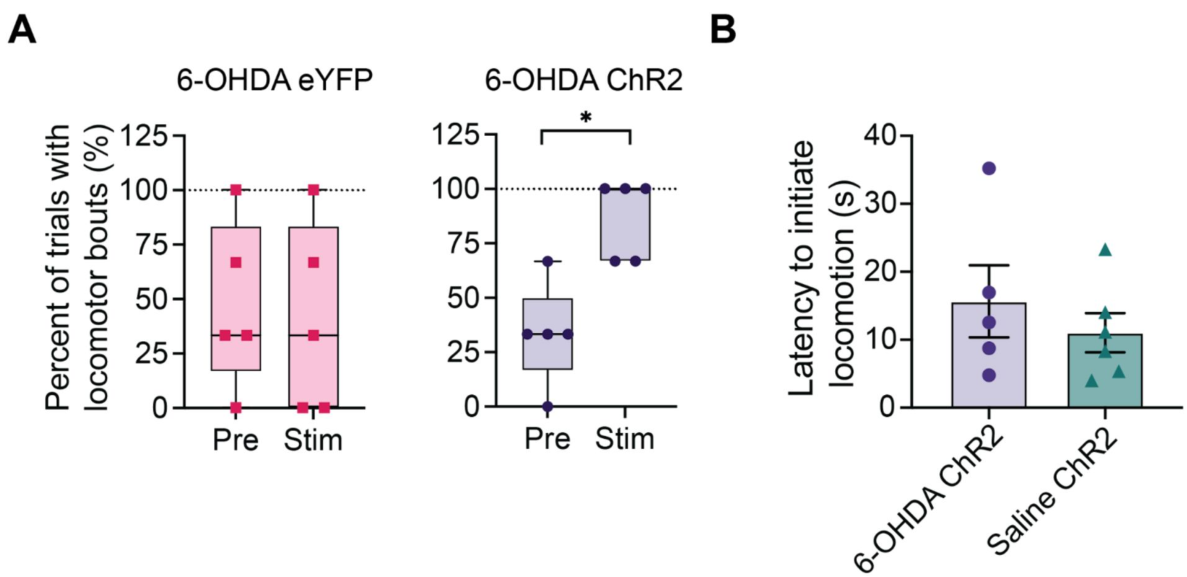


Figure S3.

Characterization of A13 region photoactivation temporal dynamics on locomotion initiation.

(A) Percent of trials where there was at least one bout of locomotion. Data are plotted as box and whiskers with the horizontal line through the box indicating the group median, interquartile range indicated by the limits of the box, and group minimum and maximum indicated by the whiskers. Asterisks indicate significant comparisons using the Wilcoxon signed-rank test: * $p < 0.05$. **(B)** The average time for the ChR2 group animals to begin locomotion after the onset of photoactivation was not different from sham mice ($p = 0.953$). Means plotted with error bars indicating ± SEM.

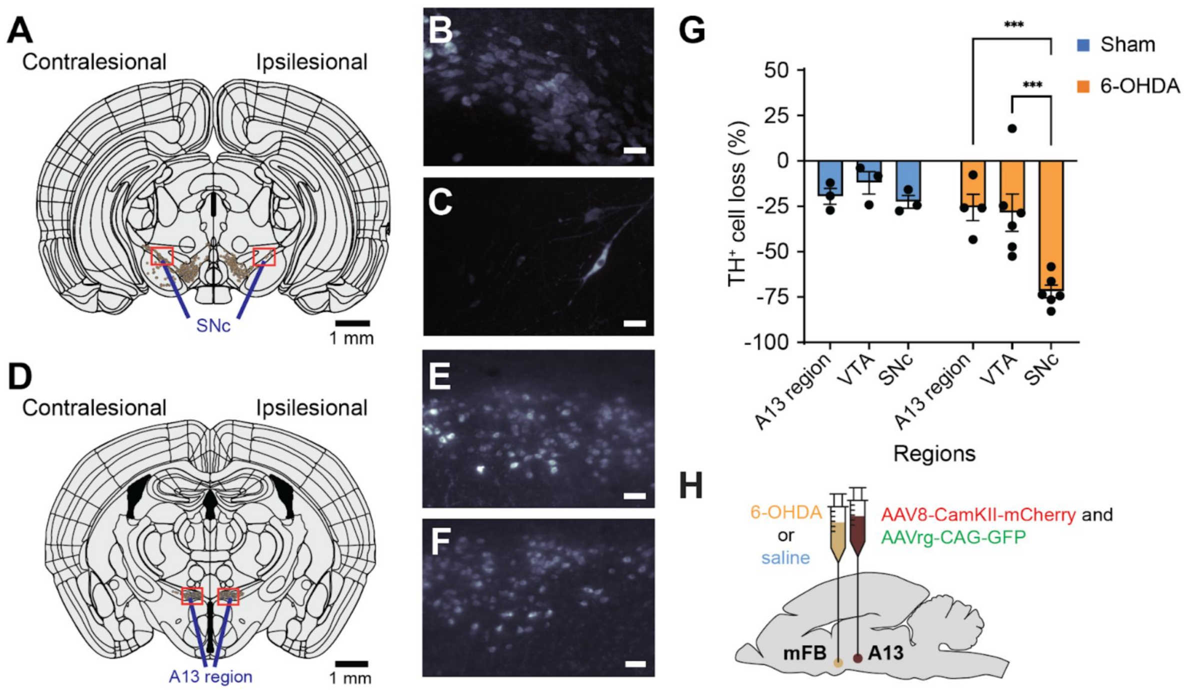


Figure S4.

Preservation of TH+ A13 cells in Parkinsonian mouse models.

Representative slices of SNc (AP: -3.08mm, **A**) and A13 region (AP: -1.355mm, **D**) following registration with WholeBrain software. Full 3D brain is available (see Movie S4). There was a lack of TH^+ SNc cells following 6-OHDA injections at the MFB (**A**). (**B, C**) Zoomed sections (90 µm thickness) of red boxes in panel A in left to right order. Meanwhile, TH^+ VTA cells were preserved bilaterally. In addition, TH^+ A13 cells were present ipsilesional to 6-OHDA injections (**D**). (**E, F**) Zoomed sections (90 µm thickness) of red boxes in panel D in left to right order. When calculating the percentage of TH^+ cell loss normalized to the intact side, there was a significant interaction between the condition group and brain regions (repeated measures two-way ANOVA with *post hoc* Bonferroni pairwise, sham: $n = 3$, 6-OHDA: $n = 6$) (**G**) 6-OHDA-treated mice showed a significantly greater percentage of TH^+ cell loss in SNc compared to VTA and A13 region (VTA vs. SNc: $p = 0.005$; A13region vs. SNc: $p < 0.05$). In contrast, sham showed no significant difference in TH^+ cell loss across SNc, VTA and A13 region regions ($p > 0.05$). $*p < 0.05$, and $**p < 0.01$. (H) Dual ipsilateral stereotaxic injection into the MFB and A13 region. Scale bars are 50 µm unless otherwise indicated.

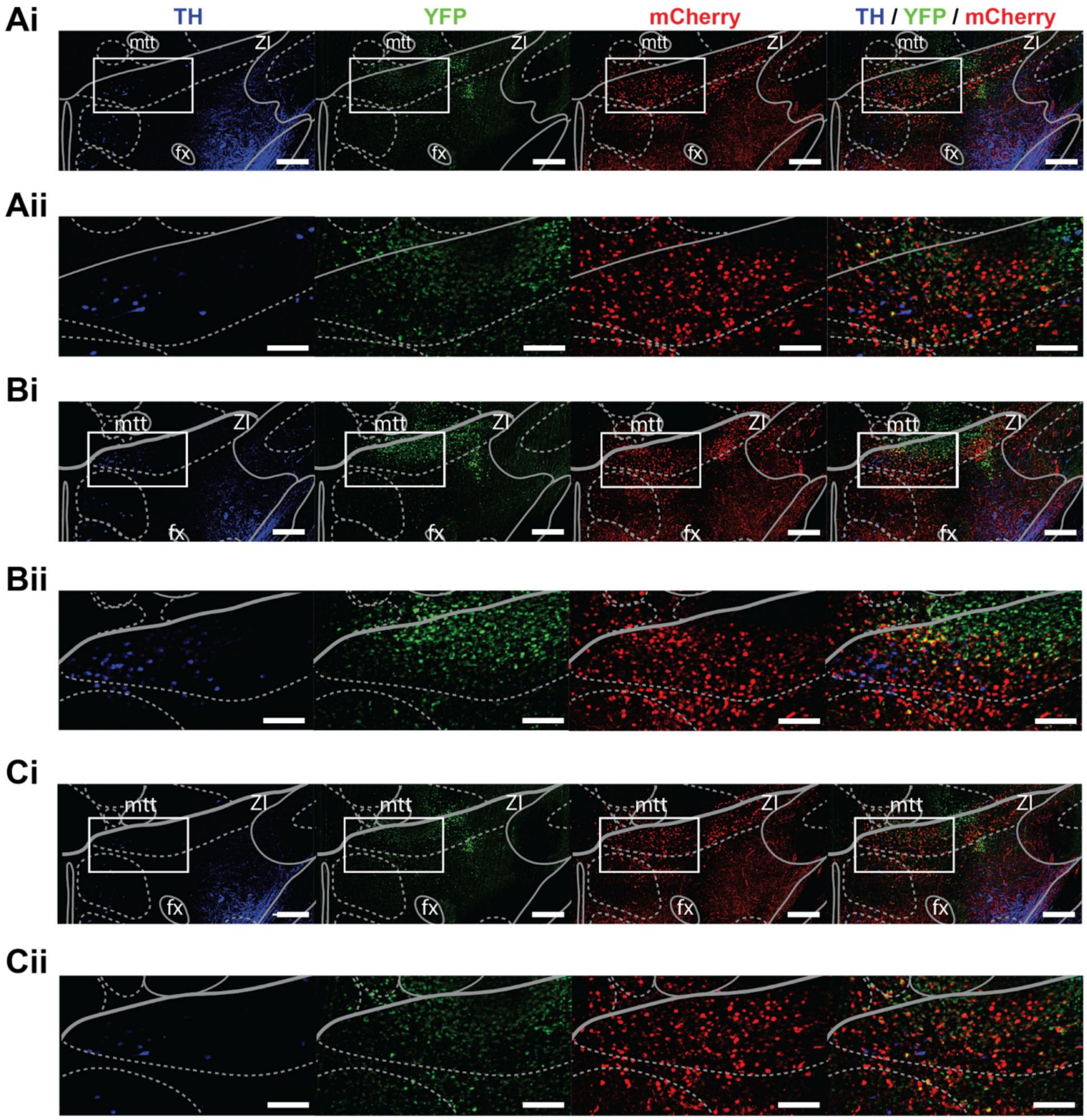


Figure S5.

Example of the injection core in a sham brain for viral tracers and the rostral and caudal spread to the injection site (A13).

Viral tracers (AAV8-CamKII-mCherry and AAVrg-CAG-GFP) were mixed 50:50. Light-sheet images around the injection site were obtained with 2x objective, 6.3x optical zoom, and a z-step size of 2 μm (xyz resolution = 0.477 μm x 0.477 μm x 2 μm). Background filtering (median value of 20 pixels and Gaussian smoothing with a sigma value of 10) was performed in ImageJ software 1 and visualized in IMARIS 9.8 (Belfast, United Kingdom). 2008 Allen reference atlas images were overlaid on top of 90 μm maximum intensity projections taken from IMARIS 9.8 (Belfast, United Kingdom): -1.26 mm **(A)**, -1.36 mm **(B)**, and -1.46 mm **(C)**. Zoomed in sections of each white rectangular area at each coordinate (rows '**i**') are displayed below for each fluorophore (rows '**ii**'). Scale bars for rows 'i' are 200 μm and for rows 'ii are 100 μm.

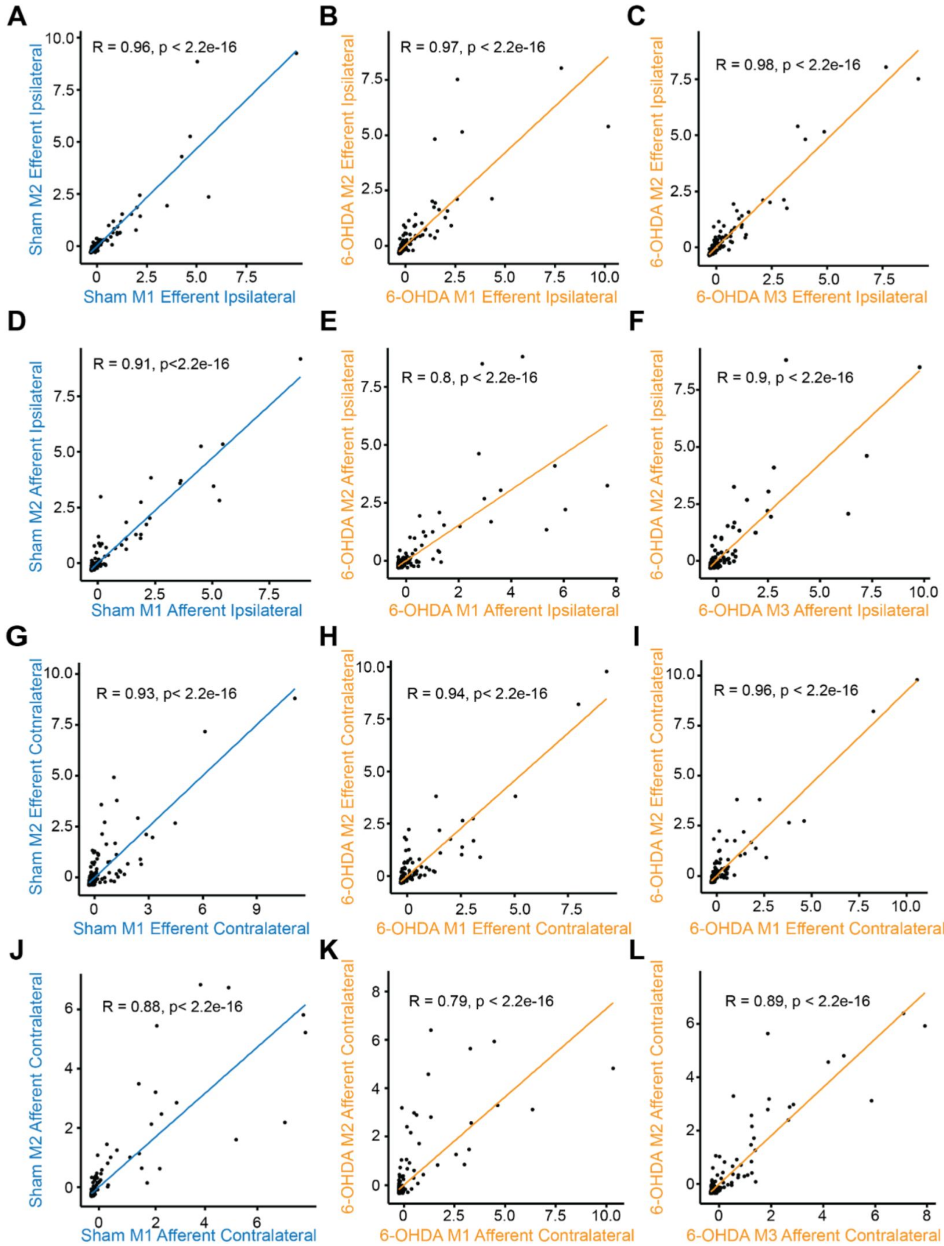


Figure S6.

The ipsilateral (A-F) and contralateral (G-L) afferent and efferent proportions across Sham (blue) and 6-OHDA (orange) mice were consistent when compared in a pairwise manner. An experimental variation on the total labeling of neurons and fibers was minimized by dividing the afferent cell counts or efferent fiber areas in each brain region by the total number found in a brain to obtain the proportion of total inputs and outputs. Using Spearman's correlation analysis, we found afferent and efferent proportions across animals to be consistent among each other with an average correlation of 0.91 (SEM = 0.02). M1 = mouse #1, M2 = mouse #2, M3 = mouse #3.

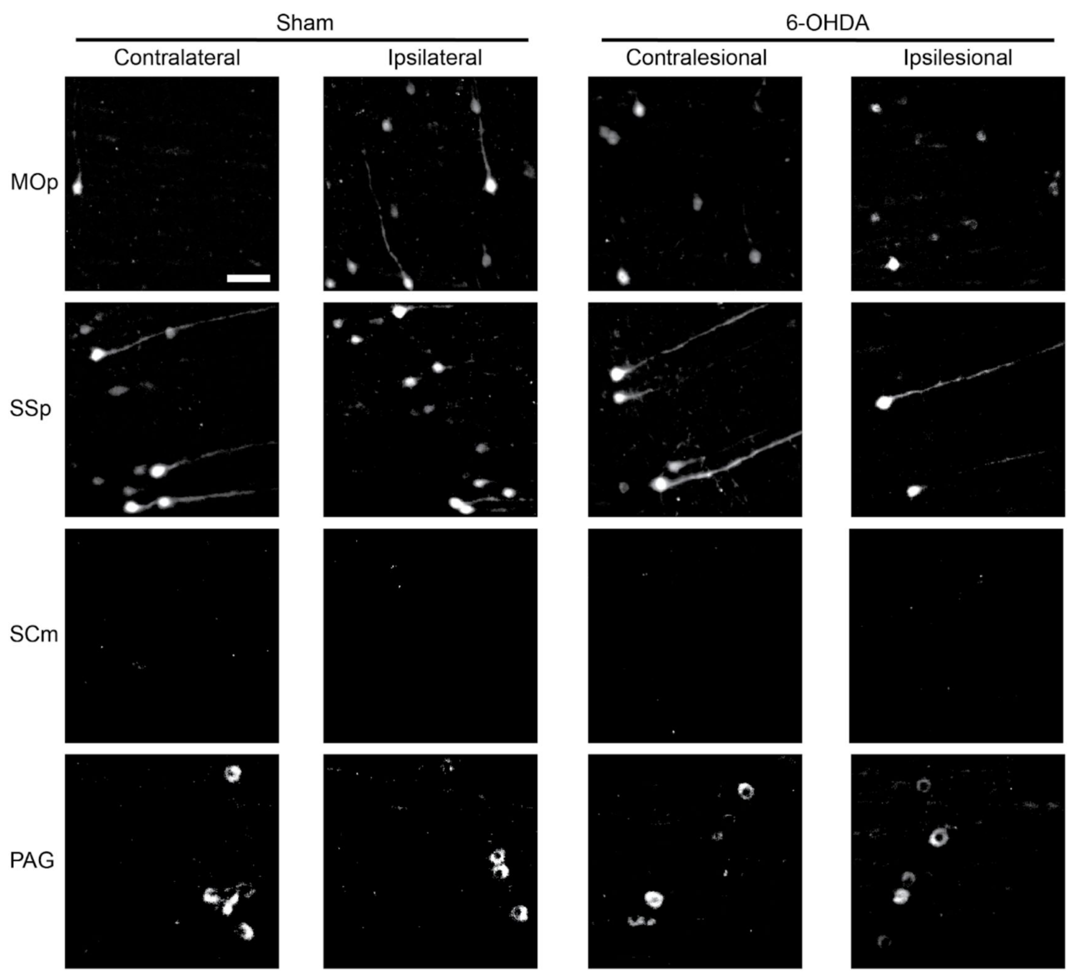


Figure S7.

Examples of retrogradely labelled GFP positive fibers and cells from selected regions illustrating cell bodies from whole brain imaging that project to the A13 region. Expressions of GFP detected using light-sheet microscopy with 2x objective, 6.3x optical zoom, and a z-step size of 2 μm (xyz resolution = 0.477 μm x 0.477 μm x 2 μm). Brain regions were delimited by registration with Allen Brain Atlas (see Methods) and cropped out of selected 90 μm image stack in ImageJ software 1. A background filter (Gaussian smoothing with a rolling ball radius of 20 pixels) and a minimum filter with radius of 1 pixel were performed in ImageJ software 1. Scale bar = 50 μm.

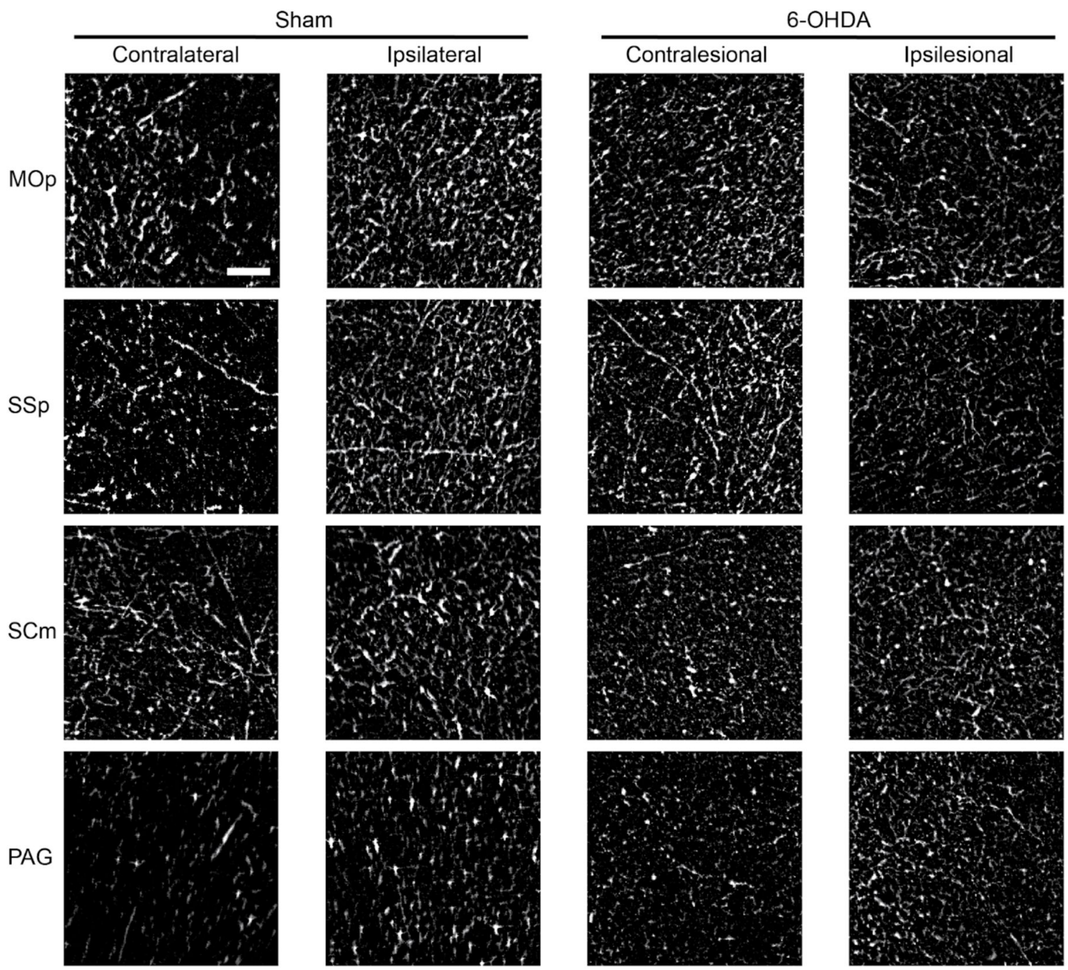


Figure S8.

Examples of anterogradely labelled mCherry positive fibers from selected regions illustrating cell bodies from whole brain imaging that project to the A13 region. Expressions of mCherry detected using light-sheet microscopy with 2x objective, 6.3x optical zoom, and a z-step size of 2 µm (xyz resolution = 0.477 µm x 0.477 µm x 2 µm). Brain regions were delimited by registration with Allen Brain Atlas (see Methods) and cropped out of selected 90 µm image stack in ImageJ software 1. A background filter (Gaussian smoothing with a rolling ball radius of 5 pixels) and a minimum filter with a radius of 1 pixel were applied in ImageJ software 1. Scale bar = 50 µm.

Data availability

All datasets and code will be available in a public repository.

Author Contributions

LHK and APL performed experiments and prepared figures. CB and MAT prepared and edited figures. PJW edited figures and procured funding for the experiments. LHK, APL, ZHTK and PJW conceived and designed the research and interpreted the results. SS and APL performed surgeries for lesions, optogenetic experiments and conducted behavioral experiments. LHK and SEAE optimized light-sheet imaging. MAT, CB, ST, and CM performed manual cell counting. TES performed analysis. TC performed analysis and prepared figures on gait analysis. LHK, APL, TC, ZHTK, and PJW drafted the manuscript. All authors reviewed and approved the final version of the manuscript.

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Additional files

Movie S1. *Photoactivation of the A13 region in a 6-OHDA model mouse producing increased locomotion in the OFT (2x speed).*

Movie S2. *Photoactivation of the A13 region in a sham mouse producing increased locomotion in the OFT (2x speed).*

Movie S3. *Photoactivation of the A13 region during the pole test in a 6-OHDA model mouse decreases pole descent time (0.5x speed).*

Movie S4. *Video showing TH staining following whole brain imaging and staining in a 6-OHDA model mouse brain. Focus on TH expression in the A13 and SNc regions.*

References

1. Berg S, Kutra D, Kroeger T, Straehle CN, Kausler BX, Haubold C, Schiegg M, Ales J, Beier T, Rudy M, Eren K, Cervantes JI, Xu B, Beuttenmueller F, Wolny A, Zhang C, Koethe U, Hamprecht FA, Kreshuk A (2019) **. ilastik: interactive machine learning for (bio)image analysis** *Nat Methods* **16**:1226–1232

2. Blomstedt P, Stenmark Persson R, Hariz G-M, Linder J, Fredricks A, Häggström B, Philipsson J, Forsgren L, Hariz M (2018) **Deep brain stimulation in the caudal zona incerta versus best medical treatment in patients with Parkinson's disease: a randomised blinded evaluation** *J Neurol Neurosurg Psychiatry* **89**:710–716

3. Boix J, Padel T, Paul G (2015) **A partial lesion model of Parkinson's disease in mice--characterization of a 6-OHDA-induced medial forebrain bundle lesion** *Behav Brain Res* **284**:196–206

4. Bolton AD, Murata Y, Kirchner R, Kim S-Y, Young A, Dang T, Yanagawa Y, Constantine-Paton M (2015) **A Diencephalic Dopamine Source Provides Input to the Superior Colliculus, where D1 and D2 Receptors Segregate to Distinct Functional Zones** *Cell Rep* **13**:1003–1015

5. Bosson A, Boisseau S, Buisson A, Savasta M, Albrieux M (2015) **Disruption of dopaminergic transmission remodels tripartite synapse morphology and astrocytic calcium activity within substantia nigra pars reticulata** *Glia* **63**:673–683

6. Braak H, Del Tredici K, Rüb U, de Vos RAI, Jansen Steur ENH, Braak E (2003) **Staging of brain pathology related to sporadic Parkinson's disease** *Neurobiol Aging* **24**:197–211

7. Caire F, Ranoux D, Guehl D, Burbaud P, Cuny E (2013) **A systematic review of studies on anatomical position of electrode contacts used for chronic subthalamic stimulation in Parkinson's disease** *Acta Neurochir* **155**:1647–54

8. Carey RJ (1991) **Chronic L-dopa treatment in the unilateral 6-OHDA rat: evidence for behavioral sensitization and biochemical tolerance** *Brain Res* **568**:205–214

9. Cenci MA, Björklund A (2020) **Animal models for preclinical Parkinson's research: An update and critical appraisal** *Prog Brain Res* **252**:27–59

10. Chen F, Qian J, Cao Z, Li A, Cui J, Shi L, Xie J (2023) **Chemogenetic and optogenetic stimulation of zona incerta GABAergic neurons ameliorates motor impairment in Parkinson's disease** *iScience* **26** https://doi.org/10.1016/j.isci.2023.107149

11. Choi EA, McNally GP (2017) **Paraventricular Thalamus Balances Danger and Reward** *J Neurosci* **37**:3018–3029

12. Chometton S, Charrière K, Bayer L, Houdayer C, Franchi G, Poncet F, Fellmann D, Risold PY (2017) **The rostromedial zona incerta is involved in attentional processes while adjacent LHA responds to arousal: c-Fos and anatomical evidence** *Brain Struct Funct* **222**:2507–2525

13. Cohen J (1988) **Statistical power analysis for the behavioral sciences New York** NY: Academic

14. Costall B, Naylor RJ, Pycock C (1976) **Non-specific supersensitivity of striatal dopamine receptors after 6-hydroxydopamine lesion of the nigrostriatal pathway** *Eur J Pharmacol* **35**:276–283

15. Dauer W, Przedborski S (2003) **Parkinson's disease: mechanisms and models** *Neuron* **39**:889–909

16. DeLong MR, Wichmann T (2015) **Basal Ganglia Circuits as Targets for Neuromodulation in Parkinson Disease** *JAMA Neurol* **72**:1354–1360

17. Eaton MJ, Wagner CK, Moore KE, Lookingland KJ (1994) **Neurochemical identification of A13 dopaminergic neuronal projections from the medial zona incerta to the horizontal limb of the diagonal band of Broca and the central nucleus of the amygdala** *Brain Res* **659**:201–207

18. Ferraye MU, Debû B, Fraix V, Goetz L, Ardouin C, Yelnik J, Henry-Lagrange C, Seigneuret E, Piallat B, Krack P, Le Bas J-F, Benabid A-L, Chabardès S, Pollak P (2010) **Effects of pedunculopontine nucleus area stimulation on gait disorders in Parkinson's disease** *Brain* **133**:205–214

19. Fischer P, He S, de Roquemaurel A, Akram H, Foltynie T, Limousin P, Zrinzo L, Hyam J, Cagnan H, Brown P, Tan H (2020) **Entraining Stepping Movements of Parkinson's Patients to Alternating Subthalamic Nucleus Deep Brain Stimulation** *J Neurosci* **40**:8964–8972

20. Fougère M, van der Zouwen CI, Boutin J, Neszvecsko K, Sarret P, Ryczko D (2021) **Optogenetic stimulation of glutamatergic neurons in the cuneiform nucleus controls locomotion in a mouse model of Parkinson's disease** *Proc Natl Acad Sci U S A* **118** https://doi.org/10.1073/pnas.2110934118

21. Fürth D, Vaissière T, Tzortzi O, Xuan Y, Märtin A, Lazaridis I, Spigolon G, Fisone G, Tomer R, Deisseroth K, Carlén M, Miller CA, Rumbaugh G, Meletis K (2018) **An interactive framework for whole-brain maps at cellular resolution** *Nat Neurosci* **21**:139–149

22. Garau C, Hayes J, Chiacchierini G, McCutcheon JE, Apergis-Schoute J (2023) **Involvement of A13 dopaminergic neurons in prehensile movements but not reward in the rat** *Curr Biol* https://doi.org/10.1016/j.cub.2023.09.044

23. Glajch KE, Fleming SM, Surmeier DJ, Osten P (2012) **Sensorimotor assessment of the unilateral 6-hydroxydopamine mouse model of Parkinson's disease** *Behav Brain Res* **230**:309–316

24. Glickstein M, Stein J (1991) **Paradoxical movement in Parkinson's disease** *Trends Neurosci* **14**:480–482

25. Goldowitz D (2010) **Allen Reference Atlas. A Digital Color Brain Atlas of the C57BL/6J Male Mouse - by H. W. Dong** *Genes, Brain and Behavior* https://doi.org/10.1111/j.1601-183x.2009.00552.x

26. Gradinaru V, Mogri M, Thompson KR, Henderson JM, Deisseroth K (2009) **Optical deconstruction of parkinsonian neural circuitry** *Science* **324**:354–359

27. Grossman RG (1958) **Effects of stimulation of non-specific thalamic system on locomotor movements in cat** *J Neurophysiol* **21**:85–93

28. Gut NK, Winn P (2015) **Deep Brain Stimulation of Different Pedunculopontine Targets in a Novel Rodent Model of Parkinsonism** *Journal of Neuroscience* https://doi.org/10.1523/jneurosci.3646-14.2015

29. Haery L, Deverman BE, Matho KS, Cetin A, Woodard K, Cepko C, Guerin KI, Rego MA, Ersing I, Bachle SM, Kamens J, Fan M (2019) **Adeno-Associated Virus Technologies and Methods for Targeted Neuronal Manipulation** *Front Neuroanat* **13**:93

30. Hamani C, Moro E, Lozano AM (2011) **The pedunculopontine nucleus as a target for deep brain stimulation** *J Neural Transm* **118**:1461–1468

31. Hansen HH, Roostalu U, Hecksher-Sørensen J (2020) **Whole-brain three-dimensional imaging for quantification of drug targets and treatment effects in mouse models of neurodegenerative diseases** *Neural Regeneration Res* **15**:2255–2257

32. Hoffman BJ, Palkovits M, Paiak K, Hamson SR, Mezey É (1997) **Regulation of Dopamine Transporter mRNA Levels in the Central Nervous System** In: Goldstein DS, Eisenhofer G, McCarty R, editors. *, editors. Advances in Pharmacology* Academic Press pp. 202–206

33. Hormigo S, Zhou J, Castro-Alamancos MA (2020) **Zona Incerta GABAergic Output Controls a Signaled Locomotor Action in the Midbrain Tegmentum** *eNeuro* **7** https://doi.org/10.1523/ENEURO.0390-19.2020

34. Hughes AJ, Ben-Shlomo Y, Daniel SE, Lees AJ (1992) **What features improve the accuracy of clinical diagnosis in Parkinson's disease: a clinicopathologic study** *Neurology* **42**:1142–1146

35. Iancu R, Mohapel P, Brundin P, Paul G (2005) **Behavioral characterization of a unilateral 6-OHDA-lesion model of Parkinson's disease in mice** *Behav Brain Res* **162**:1–10

36. Jagmag SA, Tripathi N, Shukla SD, Maiti S, Khurana S (2015) **Evaluation of Models of Parkinson's Disease** *Front Neurosci* **9**:503

37. Kimbrough A, Lurie DJ, Collazo A, Kreifeldt M, Sidhu H, Macedo GC, D'Esposito M, Contet C, George O (2020) **Brain-wide functional architecture remodeling by alcohol dependence and abstinence** *Proc Natl Acad Sci U S A* **117**:2149–2159

38. Kim L, Chomiak T, Tran MA, Tam S, McPherson C, Eaton SEA, Ou Y, Kiss ZHT, Whelan PJ (2021) **Global remodelling of afferent and efferent projections of the A13 region following unilateral nigrostriatal degeneration using 6-hydroxydopamineNeuroscience Meeting Planner** In: *Society for Neuroscience*

39. Kim LH, Chomiak T, Tran M, Tam S, McPherson C, Eaton SEA, Ou Y, Resch L, Kiss ZHT, Whelan PJ (2021) **Substantia nigra degradation results in widespread changes in medial zona incerta afferent and efferent connectomics** *bioRxiv*

40. Kim LH, Sharma S, Sharples SA, Mayr KA, Kwok CHT, Whelan PJ (2017) **Integration of Descending Command Systems for the Generation of Context-Specific Locomotor Behaviors** *Front Neurosci* **11**:581

41. Kish SJ, Tong J, Hornykiewicz O, Rajput A, Chang L-J, Guttman M, Furukawa Y (2008) **Preferential loss of serotonin markers in caudate versus putamen in Parkinson's disease** *Brain* **131**:120–131

42. Koblinger K, Jean-Xavier C, Sharma S, Füzesi T, Young L, Eaton SEA, Kwok CHT, Bains JS, Whelan PJ (2018) **Optogenetic Activation of A11 Region Increases Motor Activity** *Front Neural Circuits* **12**:86

43. Konnova EA (2018) **Animal models of Parkinson's disease** In: *Parkinson's Disease: Pathogenesis and Clinical Aspects* Codon Publications pp. 83–106

44. Lane EL, Cheetham SC, Jenner P (2006) **Does contraversive circling in the 6-OHDA-lesioned rat indicate an ability to induce motor complications as well as therapeutic effects in Parkinson's disease?** *Exp Neurol* **197**:284–290

45. Lee CS, Schulzer M, Mak E, Hammerstad JP, Calne S, Calne DB (1995) **Patterns of asymmetry do not change over the course of idiopathic parkinsonism: implications for pathogenesis** *Neurology* **45**:435–439

46. Li L-X, Li Y-L, Wu J-T, Song J-Z, Li X-M (2021) **Glutamatergic Neurons in the Caudal Zona Incerta Regulate Parkinsonian Motor Symptoms in Mice** *Neurosci Bull* https://doi.org/10.1007/s12264-021-00775-9

47. Lim S-Y, Fox SH, Lang AE (2009) **Overview of the extranigral aspects of Parkinson disease** *Arch Neurol* **66**:167–172

48. Li S, Shi Y, Kirouac GJ (2014) **The hypothalamus and periaqueductal gray are the sources of dopamine fibers in the paraventricular nucleus of the thalamus in the rat** *Front Neuroanat.* **8**:136

49. Lizarraga KJ, Jagid JR, Luca CC (2016) **Comparative effects of unilateral and bilateral subthalamic nucleus deep brain stimulation on gait kinematics in Parkinson's disease: a randomized, blinded study** *J Neurol* **263**:1652–1656

50. Li Z, Rizzi G, Tan KR (2021) **Zona incerta subpopulations differentially encode and modulate anxiety** *Sci Adv* **7**:eabf6709

51. Magno LAV, Tenza-Ferrer H, Collodetti M, Aguiar MFG, Rodrigues APC, da Silva RS, Nicolau NF, Rosa DVF, Birbrair A, Miranda DM, Romano-Silva MA (2019) **Optogenetic Stimulation of the M2 Cortex Reverts Motor Dysfunction in a Mouse Model of Parkinson's Disease** *J Neurosci* **39**:3234–3248

52. Masini D, Kiehn O (2022) **Targeted activation of midbrain neurons restores locomotor function in mouse models of parkinsonism** *Nat Commun* **13**:504

53. Matsuura K, Kabuto H, Makino H, Ogawa N (1997) **Pole test is a useful method for evaluating the mouse movement disorder caused by striatal dopamine depletion** *J Neurosci Methods* **73**:45–48

54. Matzuk MM, Saper CB (1985) **Preservation of hypothalamic dopaminergic neurons in Parkinson's disease** *Ann Neurol* **18**:552–555

55. Meredith GE, Rademacher DJ (2011) **MPTP mouse models of Parkinson's disease: an update** *J Parkinsons Dis* **1**:19–33

56. Messanvi F, Eggens-Meijer E, Roozendaal B, van der Want JJ (2013) **A discrete dopaminergic projection from the incertohypothalamic A13 cell group to the dorsolateral periaqueductal gray in rat** *Front Neuroanat* **7**:41

57. Mitrofanis J (2005) **Some certainty for the "zone of uncertainty"? Exploring the function of the zona incerta** *Neuroscience* **130**:1–15

58. Mitrofanis J, Mikuletic L (1999) **Organisation of the cortical projection to the zona incerta of the thalamus** *J Comp Neurol* **412**:173–185

59. Mok D, Mogenson GJ (1986) **Contribution of zona incerta to osmotically induced drinking in rats** *Am J Physiol* **251**:R823–32

60. Monosov IE, Ogasawara T, Haber SN, Heimel JA, Ahmadlou M (2022) **The zona incerta in control of novelty seeking and investigation across species** *Curr Opin Neurobiol* **77**:102650

61. Moriya S, Yamashita A, Masukawa D, Setoyama H, Hwang Y, Yamanaka A, Kuwaki T (2020) **Involvement of A13 dopaminergic neurons located in the zona incerta in nociceptive processing: a fiber photometry study** *Mol Brain* **13**:60

62. Moro E, Hamani C, Poon Y-Y, Al-Khairallah T, Dostrovsky JO, Hutchison WD, Lozano AM (2010) **Unilateral pedunculopontine stimulation improves falls in Parkinson's disease** *Brain* **133**:215–224

63. Negishi K, Payant MA, Schumacker KS, Wittmann G, Butler RM, Lechan RM, Steinbusch HWM, Khan AM, Chee MJ (2020) **Distributions of hypothalamic neuron populations coexpressing tyrosine hydroxylase and the vesicular GABA transporter in the mouse** *J Comp Neurol* **528**:1833–1855

64. Neumann W-J, Steiner LA, Milosevic L (2023) **Neurophysiological mechanisms of deep brain stimulation across spatiotemporal resolutions** *Brain* **146**:4456–4468

65. Noga BR, Whelan PJ (2022) **The Mesencephalic Locomotor Region: Beyond Locomotor Control** *Front Neural Circuits* **16**:884785

66. Nonnekes J, Bereau M, Bloem BR (2020) **Freezing of Gait and Its Levodopa Paradox** *JAMA Neurol* **77**:287–288

67. Nonnekes J, Snijders AH, Nutt JG, Deuschl G, Giladi N, Bloem BR (2015) **Freezing of gait: a practical approach to management** *Lancet Neurol* **14**:768–778

68. Ogawa N, Hirose Y, Ohara S, Ono T, Watanabe Y (1985) **A simple quantitative bradykinesia test in MPTP-treated mice** *Res Commun Chem Pathol Pharmacol* **50**:435–441

69. Ogundele OM, Lee CC, Francis J (2017) **Thalamic dopaminergic neurons projects to the paraventricular nucleus-rostral ventrolateral medulla/C1 neural circuit** *The Anatomical Record* https://doi.org/10.1002/ar.23528

70. Okun MS, Foote KD (2010) **Parkinson's disease DBS: what, when, who and why? The time has come to tailor DBS targets** *Expert Rev Neurother* **10**:1847–1857

71. Ossowska K (2019) **Zona incerta as a therapeutic target in Parkinson's disease** *Journal of Neurology* https://doi.org/10.1007/s00415-019-09486-8

72. Paxinos G, Franklin KBJ (2008) **The mouse brain in stereotaxic coordinates, compact: The coronal plates and diagrams, 3rd ed** San Diego, CA: Academic Press

73. Peoples C, Spana S, Ashkan K, Benabid A-L, Stone J, Baker GE, Mitrofanis J (2012) **Photobiomodulation enhances nigral dopaminergic cell survival in a chronic MPTP mouse model of Parkinson's disease** *Parkinsonism Relat Disord* **18**:469–476

74. Perez-Lloret S, Barrantes FJ (2016) **Deficits in cholinergic neurotransmission and their clinical correlates in Parkinson's disease** *NPJ Parkinsons Dis* **2**:16001

75. Périer C, Tremblay L, Féger J, Hirsch EC (2002) **Behavioral consequences of bicuculline injection in the subthalamic nucleus and the zona incerta in rat** *J Neurosci* **22**:8711–8719

76. Périer C, Vila M, Féger J, Agid Y, Hirsch EC (2000) **Functional activity of zona incerta neurons is altered after nigrostriatal denervation in hemiparkinsonian rats** *Exp Neurol* **162**:215–224

77. Perović M, Mladenović A, Rakić L, Ruzdijić S, Kanazir S (2005) **Increase of GAP-43 in the rat cerebellum following unilateral striatal 6-OHDA lesion** *Synapse* **56**:170–174

78. Plaha P, Khan S, Gill SS (2008) **Bilateral stimulation of the caudal zona incerta nucleus for tremor control** *J Neurol Neurosurg Psychiatry* **79**:504–513

79. Power BD, Mitrofanis J (1999) **Evidence for extensive inter-connections within the zona incerta in rats** *Neurosci Lett* **267**:9–12

80. Renier N, Wu Z, Simon DJ, Yang J, Ariel P, Tessier-Lavigne M (2014) **iDISCO: a simple, rapid method to immunolabel large tissue samples for volume imaging** *Cell* **159**:896–910

81. Requejo C, López-de-Ipiña K, Ruiz-Ortega JÁ, Fernández E, Calvo PM, Morera-Herreras T, Miguelez C, Cardona-Grifoll L, Cepeda H, Ugedo L, Lafuente JV (2020) **Changes in Day/Night Activity in the 6-OHDA-Induced Experimental Model of Parkinson's Disease: Exploring Prodromal Biomarkers** *Front Neurosci* **14**:590029

82. Romanov RA, Zeisel A, Bakker J, Girach F, Hellysaz A, Tomer R, Alpár A, Mulder J, Clotman F, Keimpema E, Hsueh B, Crow AK, Martens H, Schwindling C, Calvigioni D, Bains JS, Máté Z, Szabó G, Yanagawa Y, Zhang M-D, Rendeiro A, Farlik M, Uhlén M, Wulff P, Bock C, Broberger C, Deisseroth K, Hökfelt T, Linnarsson S, Horvath TL, Harkany T (2017) **Molecular interrogation of hypothalamic organization reveals distinct dopamine neuronal subtypes** *Nat Neurosci* **20**:176–188

83. Roostalu U, Salinas CBG, Thorbek DD, Skytte JL, Fabricius K, Barkholt P, John LM, Jurtz VI, Knudsen LB, Jelsing J, Vrang N, Hansen HH, Hecksher-Sørensen J (2019) **Quantitative whole-brain 3D imaging of tyrosine hydroxylase-labeled neuron architecture in the mouse MPTP model of Parkinson's disease** *Dis Model Mech* **12** https://doi.org/10.1242/dmm.042200

84. Ryczko D, Cone JJ, Alpert MH, Goetz L, Auclair F, Dubé C, Parent M, Roitman MF, Alford S, Dubuc R (2016) **A descending dopamine pathway conserved from basal vertebrates to mammals** *Proc Natl Acad Sci U S A* **113**:E2440–9

85. Sahel J-A, Boulanger-Scemama E, Pagot C, Arleo A, Galluppi F, Martel JN, Delaux A, de Saint Aubert J-B, de Montleau C, Gutman E, Audo I, Duebel J, Picaud S, Dalkara D, Blouin L, Taiel M, Roska B (2021) **Partial recovery of visual function in a blind patient after optogenetic therapy** *Nat Med* **27**:1223–1229

86. Sanders TH, Jaeger D (2016) **Optogenetic stimulation of cortico-subthalamic projections is sufficient to ameliorate bradykinesia in 6-ohda lesioned mice** *Neurobiol Dis* **95**:225–237

87. Sanghera MK, Anselmo-Franci J, McCann SM (1991) **Effect of Medial Zona Incerta Lesions on the Ovulatory Surge of Gonadotrophins and Prolactin in the Rat** *Neuroendocrinology* https://doi.org/10.1159/000125931

88. Sanghera MK, Grady S, Smith W, Woodward DJ, Porter JC (1991) **Incertohypothalamic A13 dopamine neurons: effect of gonadal steroids on tyrosine hydroxylase** *Neuroendocrinology* **53**:268–275

89. Scatton B, Javoy-Agid F, Rouquier L, Dubois B, Agid Y (1983) **Reduction of cortical dopamine, noradrenaline, serotonin and their metabolites in Parkinson's disease** *Brain Res* **275**:321–328

90. Schneider CA, Rasband WS, Eliceiri KW (2012) **NIH Image to ImageJ: 25 years of image analysis** *Nat Methods* **9**:671–675

91. Schwarting RK, Bonatz AE, Carey RJ, Huston JP (1991) **Relationships between indices of behavioral asymmetries and neurochemical changes following mesencephalic 6-hydroxydopamine injections** *Brain Res* **554**:46–55

92. Sharma S, Badenhorst CA, Ashby DM, Di Vito SA, Tran MA, Ghavasieh Z, Grewal GK, Belway CR, McGirr A, Whelan PJ (2024) **Inhibitory medial zona incerta pathway drives exploratory behavior by inhibiting glutamatergic cuneiform neurons** *Nat Commun* **15**:1160

93. Sharma S, Kim LH, Mayr KA, Elliott DA, Whelan PJ (2018) **Parallel descending dopaminergic connectivity of A13 cells to the brainstem locomotor centers** *Sci Rep* **8**:7972

94. Shaw VE, Spana S, Ashkan K, Benabid A-L, Stone J, Baker GE, Mitrofanis J (2010) **Neuroprotection of midbrain dopaminergic cells in MPTP-treated mice after near-infrared light treatment** *J Comp Neurol* **518**:25–40

95. Simola N, Morelli M, Carta AR (2007) **The 6-hydroxydopamine model of Parkinson's disease** *Neurotox Res* **11**:151–167

96. Sita LV, Elias CF, Bittencourt JC (2007) **Connectivity pattern suggests that incerto-hypothalamic area belongs to the medial hypothalamic system** *Neuroscience* **148**:949–969

97. Spix TA, Nanivadekar S, Toong N, Kaplow IM, Isett BR, Goksen Y, Pfenning AR, Gittis AH (2021) **Population-specific neuromodulation prolongs therapeutic benefits of deep brain stimulation** *Science* **374**:201–206

98. (no date) **Stanford Optogenetics Resource Center. https://web.stanford.edu/group/dlab/cgi-bin/graph/chart.php no date** https://web.stanford.edu/group/dlab/cgi-bin/graph/chart.php

99. Stefani A, Lozano AM, Peppe A, Stanzione P, Galati S, Tropepi D, Pierantozzi M, Brusa L, Scarnati E, Mazzone P (2007) **Bilateral deep brain stimulation of the pedunculopontine and subthalamic nuclei in severe Parkinson's disease** *Brain* **130**:1596–1607

100. Stott SRW, Barker RA (2014) **Time course of dopamine neuron loss and glial response in the 6-OHDA striatal mouse model of Parkinson's disease** *Eur J Neurosci* **39**:1042–1056

101. Thevathasan W, Debu B, Aziz T, Bloem BR, Blahak C, Butson C, Czernecki V, Foltynie T, Fraix V, Grabli D, Joint C, Lozano AM, Okun MS, Ostrem J, Pavese N, Schrader C, Tai C-H, Krauss JK, Moro E, Movement Disorders Society PPN DBS Working Groupin collaboration with the World Society

for Stereotactic and Functional Neurosurgery (2018) **Pedunculopontine nucleus deep brain stimulation in Parkinson's disease: A clinical review** *Mov Disord* **33**::10–20

102. Thiele SL, Warre R, Nash JE (2012) **Development of a unilaterally-lesioned 6-OHDA mouse model of Parkinson's disease** *J Vis Exp* https://doi.org/10.3791/3234

103. Tsang E, Orlandini C, Sureka R, Crevenna AH, Perlas E, Prankerd I, Masferrer ME, Gross CT (2021) **Induction of flight via midbrain projections to the cuneiform nucleus** *bioRxiv* https://doi.org/10.1101/2021.12.21.473683

104. Ungerstedt U (1971) **Striatal dopamine release after amphetamine or nerve degeneration revealed by rotational behaviour** *Acta Physiol Scand Suppl* **367**:49–68

105. Valverde S, Vandecasteele M, Piette C, Derousseaux W, Gangarossa G, Aristieta Arbelaiz A, Touboul J, Degos B, Venance L (2020) **Deep brain stimulation-guided optogenetic rescue of parkinsonian symptoms** *Nat Commun* **11**:2388

106. Venkataraman A, Hunter SC, Dhinojwala M, Ghebrezadik D, Guo J, Inoue K, Young LJ, Dias BG (2021) **Incerto-thalamic modulation of fear via GABA and dopamine** *Neuropsychopharmacology* **46**:1658–1668

107. Wang X, Chou X-L, Zhang LI, Tao HW (2020) **Zona Incerta: An Integrative Node for Global Behavioral Modulation** *Trends Neurosci* **43**:82–87

108. Wardas J, Ossowska K, Wolfarth S (1988) **Evidence for the independent role of GABA synapses of the zona incerta-lateral hypothalamic region in haloperidol-induced catalepsy** *Brain Res* **462**:378–382

109. Watakabe A, Ohtsuka M, Kinoshita M, Takaji M, Isa K, Mizukami H, Ozawa K, Isa T, Yamamori T (2015) **Comparative analyses of adeno-associated viral vector serotypes 1, 2, 5, 8 and 9 in marmoset, mouse and macaque cerebral cortex** *Neurosci Res* **93**:144–157

110. Watanabe H, Sano H, Chiken S, Kobayashi K, Fukata Y, Fukata M, Mushiake H, Nambu A (2020) **Forelimb movements evoked by optogenetic stimulation of the macaque motor cortex** *Nat Commun* **11**:3253

111. Yang Y, Jiang T, Jia X, Yuan J, Li X, Gong H (2022) **Whole-Brain Connectome of GABAergic Neurons in the Mouse Zona Incerta** *Neurosci Bull* **38**:1315–1329

112. Ye Q, Nunez J, Zhang X (2023) **Zona incerta dopamine neurons encode motivational vigor in food seeking** *Sci Adv* **9**:eadi5326

113. Zetterström T, Herrera-Marschitz M, Ungerstedt U (1986) **Simultaneous measurement of dopamine release and rotational behaviour in 6-hydroxydopamine denervated rats using intracerebral dialysis** *Brain Res* **376**:1–7

114. Zhan Y, Wu H, Liu L, Lin J, Zhang S (2021) **Organic solvent-based tissue clearing techniques and their applications** *J Biophotonics* **14**:e202000413

115. Zhao Z-D, Chen Z, Xiang X, Hu M, Xie H, Jia X, Cai F, Cui Y, Chen Z, Qian L, Liu J, Shang C, Yang Y, Ni X, Sun W, Hu J, Cao P, Li H, Shen WL (2019) **Zona incerta GABAergic neurons integrate prey-related sensory signals and induce an appetitive drive to promote hunting** *Nat Neurosci* **22**:921–932

116. Zweig RM, Jankel WR, Hedreen JC, Mayeux R, Price DL (1989) **The pedunculopontine nucleus in Parkinson's disease** *Ann Neurol* **26**:41–46

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Reviewer #1 (Public review):

Summary:

This study aimed to investigate the effects of optically stimulating the A13 region in healthy mice and a unilateral 6-OHDA mouse model of Parkinson's disease (PD). The primary objectives were to assess changes in locomotion, motor behaviors, and the neural connectome. For this, the authors examined the dopaminergic loss induced by 6-OHDA lesioning. They found a significant loss of tyrosine hydroxylase (TH+) neurons in the substantia nigra pars compacta (SNc) while the dopaminergic cells in the A13 region were largely preserved. Then, they optically stimulated the A13 region using a viral vector to deliver the channelrhodopsine (CamKII promoter). In both sham and PD model mice, optogenetic stimulation of the A13 region induced pro-locomotor effects, including increased locomotion, more locomotion bouts, longer durations of locomotion, and higher movement speeds. Additionally, PD model mice exhibited increased ipsilesional turning during A13 region photoactivation. Lastly, the authors used whole-brain imaging to explore changes in the A13 region's connectome after 6-OHDA lesions. These alterations involved a complex rewiring of neural circuits, impacting both afferent and efferent projections. In summary, this study unveiled the pro-locomotor effects of A13 region photoactivation in both healthy and PD model mice. The study also indicates the preservation of A13 dopaminergic cells and the anatomical changes in neural circuitry following PD-like lesions that represent the anatomical substrate for a parallel motor pathway.

Strengths:

These findings hold significant relevance for the field of motor control, providing valuable insights into the organization of the motor system in mammals. Additionally, they offer potential avenues for addressing motor deficits in Parkinson's disease (PD). The study fills a crucial knowledge gap, underscoring its importance, and the results bolster its clinical relevance and overall strength.

The authors adeptly set the stage for their research by framing the central questions in the introduction, and they provide thoughtful interpretations of the data in the discussion section. The results section, while straightforward, effectively supports the study's primary conclusion-the pro-locomotor effects of A13 region stimulation, both in normal motor control and in the 6-OHDA model of brain damage.

Weaknesses:

(1) Anatomical investigation. I have a major concern regarding the anatomical investigation of plastic changes in the A13 connectome (Figures 4 and 5). While the methodology employed to assess the connectome is technically advanced and powerful, the results lack mechanistic insight at the cell or circuit level into the pro-locomotor effects of A13 region stimulation in both physiological and pathological conditions. This concern is exacerbated by a textual description of results that doesn't pinpoint precise brain areas or subareas but instead references large brain portions like the cortical plate, making it challenging to discern the implications for A13 stimulation. Lastly, the study is generally well-written with a smooth and straightforward style, but the connectome section presents challenges in readability and comprehension. The presentation of results, particularly the correlation matrices and correlation strength, doesn't facilitate biological understanding. It would be beneficial to explore specific pathways responsible for driving the locomotor effects of A13 stimulation, including examining the strength of connections to well-known locomotor-associated regions like the Pedunculopontine nucleus, Cuneiformis nucleus, LPGi, and others in the diencephalon, midbrain, pons, and medulla. Additionally, identifying the primary inputs to A13 associated with motor function would enhance the study's clarity and relevance.

The study raises intriguing questions about compensatory mechanisms in Parkinson's disease a new perspective with the preservation of dopaminergic cells in A13, despite the SNc degeneration, and the plastic changes to input/output matrices. To gain inspiration for a more straightforward reanalysis and discussion of the results, I recommend the authors refer to the paper titled "Specific populations of basal ganglia output neurons target distinct brain stem areas while collateralizing throughout the diencephalon from the David Kleinfeld laboratory." This could guide the authors in investigating motor pathways across different brain regions.

(2) Description of locomotor performance. Figure 3 provides valuable data on the locomotor effects of A13 region photoactivation in both control and 6-OHDA mice. However, a more detailed analysis of the changes in locomotion during stimulation would enhance our understanding of the pro-locomotor effects, especially in the context of 6-OHDA lesions. For example, it would be informative to explore whether the probability of locomotion changes during stimulation in the control and 6-OHDA groups. Investigating reaction time, speed, total distance, and even kinematic aspects during stimulation could reveal how A13 is influencing locomotion, particularly after 6-OHDA lesions. The laboratory of Whelan has a deep knowledge of locomotion and the neural circuits driving it so these features may be instructive to infer insights on the neural circuits driving movement. On the same line, examining features like the frequency or power of stimulation related to walking patterns may help elucidate whether A13 is engaging with the Mesencephalic Locomotor Region (MLR) to drive the pro-locomotor effects. These insights would provide a more comprehensive understanding of the mechanisms underlying A13-mediated locomotor changes in both healthy and pathological conditions.

(3) Figure 2 indeed presents valuable information regarding the effects of A13 region photoactivation. To enhance the comprehensiveness of this figure and gain a deeper understanding of the neurons driving the pro-locomotor effect of stimulation, it would be beneficial to include quantifications of various cell types:

• cFos-Positive Cells/TH-Positive Cells: it can help determine the impact of A13 stimulation on dopaminergic neurons and the associated pro-locomotor effect in healthy condition and especially in the context of Parkinson's disease (PD) modeling.

• cFos-Positive Cells /TH-Negative Cells: Investigating the number of TH-negative cells activated by stimulation is also important, as it may reveal non-dopaminergic neurons that play a role in locomotor responses. Identifying the location and characteristics of these TH-negative cells can provide insights into their functional significance.
Incorporating these quantifications into Figure 2 would enhance the figure's informativeness and provide a more comprehensive view of the neuronal populations involved in the locomotor effects of A13 stimulation.

(4) Referred to Figure 3. In the main text (page 5) when describing the animal with 6-OHDA the wrong panels are indicated. It is indicated in Figure 2A-E but it should be replaced with 3A-E. Please do that.

Summary of the Study after revision

The revised manuscript reflects significant efforts to improve clarity, organization, and data interpretation. The refinements in anatomical descriptions, behavioral analyses, and contextual framing have strengthened the manuscript considerably. However, the study still lacks direct causal evidence linking anatomical remodeling to behavioral improvements, and the small sample size in the anatomical analyses remains a concern. The authors have addressed many points raised in the initial review, but further acknowledgement of the exploratory nature of these findings would enhance the scientific rigor of the work.

Key Improvements in the Revision

The revised manuscript demonstrates considerable progress in clarifying data presentation, refining behavioral analyses, and improving the contextualization of anatomical findings. The restructuring of the anatomical section now provides greater precision in describing motor-related pathways, integrating terminology from the Allen Brain Atlas. The addition of new figures (Figures 4 and 5) strengthens the accessibility of these findings by illustrating key connectivity patterns more effectively. Furthermore, the correlation matrices have been adjusted to improve interpretability, ensuring that the presented data contribute meaningfully to the overall narrative of the study.

The authors have also made significant improvements in their behavioral analyses, particularly in the organization and presentation of locomotor data. Figure 3 has been revised to distinctly separate results from 6-OHDA and sham animals, providing a clearer comparison of locomotor outcomes. Additional metrics, such as reaction time, locomotion bouts, and movement speed, further enhance the granularity of the analysis, making the results more informative.

The discussion surrounding anatomical connectivity has also been strengthened. The revised manuscript now places greater emphasis on motor-related pathways and refines its analysis of A13 efferents and afferents. A newly introduced figure provides a concise summary of these connections, improving the contextualization of the anatomical data within the study's broader scope. Moreover, the authors have addressed the translational relevance of their findings by acknowledging the differences between optogenetic stimulation and deep brain stimulation (DBS). Their discussion now better situates the findings within existing literature on PD-related motor circuits, providing a more balanced perspective on the potential implications of A13 stimulation.

Remaining Concerns

Despite these substantial improvements, a number of critical concerns remain. The anatomical findings, though insightful, remain largely correlative and do not establish a causal link between structural remodeling and locomotor recovery. While the authors argue that these data will serve as a reference for future investigations, their necessity for the core conclusions of the study is not entirely clear. Additionally, while the anatomical data offer an interesting perspective on A13 connectivity, their direct relevance to the study's primary goal-demonstrating the role of A13 in locomotor recovery-remains uncertain. The authors emphasize that these data will be valuable for future research, yet their integration into the study's main narrative feels somewhat supplementary. Based on this last thought of the authors it is even more relevant another key limitation lying in the small sample size used for connectivity analyses. With only two sham and three 6-OHDA animals included, the statistical confidence in the findings is inherently limited. The absence of direct statistical comparisons between ipsilesional and contralesional projections further weakens the conclusions drawn from these anatomical studies. The authors have acknowledged that obtaining the necessary samples, acquiring the data, and analyzing them is a prolonged and resource-intensive process. While this may be a valid practical limitation, it does not justify the lack of a robust statistical approach. A more rigorous statistical framework should be employed to reinforce the findings, or alternative techniques should be considered to provide additional validation. Given these constraints, it remains unclear why the authors have not opted for standard immunohistochemistry, which could provide a complementary and more statistically accessible approach to validate the anatomical findings. Employing such an approach would not only increase the robustness of the results but also strengthen the study's impact by providing an independent confirmation of the observed structural changes.

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Reviewer #2 (Public review):

Summary:

The paper by Kim et al. investigates the potential of stimulating the dopaminergic A13 region to promote locomotor restoration in a Parkinson's mouse model. Using wild-type mice, 6-OHDA injection depletes dopaminergic neurons in the substantia nigra pars compacta, without impairing those of the A13 region and the ventral tegmentum area, as previously reported in humans. Moreover, photostimulation of presumably excitatory (CAMKIIa) neurons in the vicinity of the A13 region improves bradykinesia and akinetic symptoms after 6-OHDA injection. Whole-brain imaging with retrograde and anterograde tracers reveals that the A13 region undergoes substantial changes in the distribution of its afferents and projections after 6-OHDA injection, thus suggesting a remodeling of the A13 connectome. Whether this remodelling contributes to pro-locomotor effects of the photostimulation of the A13 region remains unknown as causality was not addressed.

Strengths:

Photostimulation of presumably excitatory (CAMKIIa) neurons in the vicinity of the A13 region promotes locomotion and locomotor recovery of wild-type mice 1 month after 6-OHDA injection in the medial forebrain bundle, thus identifying a new potential target for restoring motor functions in Parkinson's disease patients. The study also provides a description of the A13 region connectome pertaining to motor behaviors and how it changes after a dopaminergic lesion. Although there is no causal link between anatomical and behavioral data, it raises interesting questions for further studies.

Weaknesses:

Although CAMKIIa is a marker of presumably excitatory neurons and can be used as an alternative marker of dopaminergic neurons, some uncertainty remains regarding the phenotype of neurons underlying recovery of akinesia and improvement of bradykinesia.

Figure 4 is improved, but the results from the correlation analyses remain difficult to interpret, as they may reflect changes in various impaired brain regions independently of the A13 region. While the analysis offers a snapshot of correlated changes within the connectome, it does not identify which specific cell or axonal populations are actually increasing or decreasing. Although functional MRI connectome analyses are well-established, anatomical data seem less suitable for this purpose. How can one interpret correlated changes in anatomical inputs or outputs between two distinct regions?

Figure 5 is also improved, but there is room for further enhancement. As currently presented, it is difficult to distinguish the differences between the sham and 6-OHDA groups. The first column could compare afferents, while the second column could compare efferents. Given the small sample size, it would be more appropriate to present individual data rather than the mean and standard deviation.

Appraisal and impact

Although the behavioral experiments are convincing, the low number of animals in the anatomical studies is insufficient to make any relevant statistical conclusions due to extremely low statistical power.

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Reviewer #3 (Public review):

Kim, Lognon et al. present an important finding on pro-locomotor effects of optogenetic activation of the A13 region, which they identify as a dopamine-containing area of the medial zona incerta that undergoes profound remodeling in terms of afferent and efferent connectivity after administration of 6-OHDA to the MFB. The authors claim to address a model of PD-related gait dysfunction, a contentious problem that can be difficult to treat by dopaminergic medication or DBS in conventional targets. They make use of an impressive array of technologies to gain insight into the role of A13 remodeling in the 6-OHDA model of PD. The evidence provided is solid and the paper is well written, but there are several general issues that reduce the value of the paper in its current form, and a number of specific, more minor ones. Also some suggestions, that may improve the paper compared to its recent form, come to mind.

The most fundamental issue that needs to be addressed is the relation of the structural to the behavioral findings. It would be very interesting to see whether the structural heterogeneity in afferent/effects projections induced by 6-OHDA is related to the degree of symptom severity and motor improvement during A13 stimulation.

The authors provide extensive interrogation of large-scale changes in the organization of the A13 region afferent and efferent distributions. It remains unclear how many animals were included to produce Fig 4 and 5. Fig S5 suggests that only 3 animals were used, is that correct? Please provide details about the heterogeneity between animals. Please provide a table detailing how many animals were used for which experiment. Were the same animals used for several experiments?

While the authors provide evidence that photoactivation of the A13 is sufficient in driving locomotion in the OFT, this pro-locomotor effect seems to be independent of 6-OHDA induced pathophysiology. Only in the pole test do they find that there seems to be a difference between Sham vs 6-OHDA concerning effects of photoactivation of the A13. Because of these behavioral findings, optogenic activation of A13 may represent a gain of function rather than disease-specific rescue. This needs to be highlighted more explicitly in the title, abstract and conclusion.

The authors claim that A13 may be a possible target for DBS to treat gait dysfunction. However, the experimental evidence provided (in particular lack of disease-specific changes in the OFT) seem insufficient to draw such conclusions. It needs to be highlighted that optogenetic activation does not necessarily have the same effects as DBS (see the recent review from Neumann et al. in Brain: https://pubmed.ncbi.nlm.nih.gov/37450573/). This is important because ZI-DBS so far had very mixed clinical effects. The authors should provide plausible reasons for these discrepancies. Is cell-specificity, that only optogenetic interventions can achieve, necessary? Can new forms of cyclic burst DBS achieve similar specificity (Spix et al, Science 2021)? Please comment.

In a recent study, Jeon et al (Topographic connectivity and cellular profiling reveal detailed input pathways and functionally distinct cell types in the subthalamic nucleus, 2022, Cell Reports) provided evidence on the topographically graded organization of STN afferents and McElvain et al. (Specific populations of basal ganglia output neurons target distinct brain stem areas while collateralizing throughout the diencephalon, 2021, Neuron) have shown similar topographical resolution for SNr efferents. Can a similar topographical organization of efferents and afferents be derived for the A13/ ZI in total?

In conclusion, this is an interesting study that can be improved taking into consideration the points mentioned above.

https://doi.org/10.7554/eLife.90832.2.sa1

Author response:

The following is the authors’ response to the original reviews

> ***Public Reviews:***
>
> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *This study aimed to investigate the effects of optically stimulating the A13 region in healthy mice and a unilateral 6-OHDA mouse model of Parkinson's disease (PD). The primary objectives were to assess changes in locomotion, motor behaviors, and the neural connectome. For this, the authors examined the dopaminergic loss induced by 6-OHDA lesioning. They found a significant loss of tyrosine hydroxylase (TH+) neurons in the substantia nigra pars compacta (SNc) while the dopaminergic cells in the A13 region were largely preserved. Then, they optically stimulated the A13 region using a viral vector to deliver the channelrhodopsine (CamKII promoter). In both sham and PD model mice, optogenetic stimulation of the A13 region induced pro-locomotor effects, including increased locomotion, more locomotion bouts, longer durations of locomotion, and higher movement speeds. Additionally, PD model mice exhibited increased ipsi lesional turning during A13 region photoactivation. Lastly, the authors used whole-brain imaging to explore changes in the A13 region's connectome after 6-OHDA lesions. These alterations involved a complex rewiring of neural circuits, impacting both afferent and*

> *efferent projections. In summary, this study unveiled the pro-locomotor effects of A13 region photoactivation in both healthy and PD model mice. The study also indicates the preservation of A13 dopaminergic cells and the anatomical changes in neural circuitry following PD-like lesions that represent the anatomical substrate for a parallel motor pathway.*
>
> *Strengths:*
>
> *These findings hold significant relevance for the field of motor control, providing valuable insights into the organization of the motor system in mammals. Additionally, they offer potential avenues for addressing motor deficits in Parkinson's disease (PD). The study fills a crucial knowledge gap, underscoring its importance, and the results bolster its clinical relevance and overall strength.*
>
> *The authors adeptly set the stage for their research by framing the central questions in the introduction, and they provide thoughtful interpretations of the data in the discussion section. The results section, while straightforward, effectively supports the study's primary conclusion - the pro-locomotor effects of A13 region stimulation, both in normal motor control and in the 6-OHDA model of brain damage.*

We thank the reviewer for their positive comments.

> *Weaknesses:*
>
> *(1) Anatomical investigation. I have a major concern regarding the anatomical investigation of plastic changes in the A13 connectome (Figures 4 and 5). While the methodology employed to assess the connectome is technically advanced and powerful, the results lack mechanistic insight at the cell or circuit level into the pro-locomotor effects of A13 region stimulation in both physiological and pathological conditions. This concern is exacerbated by a textual description of results that doesn't pinpoint precise brain areas or subareas but instead references large brain portions like the cortical plate, making it challenging to discern the implications for A13 stimulation. Lastly, the study is generally well-written with a smooth and straightforward style, but the connectome section presents challenges in readability and comprehension. The presentation of results, particularly the correlation matrices and correlation strength, doesn't facilitate biological understanding. It would be beneficial to explore specific pathways responsible for driving the locomotor effects of A13 stimulation, including examining the strength of connections to well-known locomotor-associated regions like the Pedunculopontine nucleus, Cuneiformis nucleus, LPGi, and others in the diencephalon, midbrain, pons, and medulla.*

We initially considered two approaches. The first was to look at specific projections to the motor regions, focusing on the MLR. The second was to utilize a whole-brain analysis, which is presented here. Given what we know about the zona incerta, especially its integrative role, we felt that examining the full connectome was a reasonable starting point.

The value of the whole-brain approach is that it provides a high-level overview of the afferents and efferents to the region. The changes in the brain that occur following Parkinson-like lesions, such as those in the nigrostriatal pathway, are complex and can affect neighbouring regions such as the A13. Therefore, we wished to highlight the A13, which we considered a therapeutic target, and examine changes in connectivity that could occur following acute lesions affecting the SNc. We acknowledge that this study does not provide a causal link, but it presents the fundamental background information for subsequent hypothesis-driven, focused, region-specific analysis.

The terms provided were taken from the Allen Brain Atlas terminology and presented as abbreviations. We have added two new figures focusing on motor regions to make the information more comprehensible (new Figures 4 and 5) and rewrote the connectomics section to make it easier to understand.

> *Additionally, identifying the primary inputs to A13 associated with motor function would enhance the study's clarity and relevance.*

This is a great point to help simplify the whole-brain results. We have presented the motor-related inputs and outputs as part of a new figure in the main paper (Figure 5) and added accompanying text in the results section. We have also updated the correlation matrices to concentrate on motor regions (Figure 4). This highlights possible therapeutic pathways. We have also enhanced our discussion of these motor-related pathways. We have retained the entire dataset and added it to our data repository for those interested.

> *The study raises intriguing questions about compensatory mechanisms in Parkinson's disease and a new perspective on the preservation of dopaminergic cells in A13, despite the SNc degeneration, and the plastic changes to input/output matrices. To gain inspiration for a more straightforward reanalysis and discussion of the results, I recommend the authors refer to the paper titled "Specific populations of basal ganglia output neurons target distinct brain stem areas while collateralizing throughout the diencephalon from the David Kleinfeld laboratory." This could guide the authors in investigating motor pathways across different brain regions.*

Thank you for the advice. As pointed out, Kleinfeld's group presented their data in a nice, focused way. For the connectomic piece, we have added Figure 5, which provides a better representation than our previous submission.

> *(2) Description of locomotor performance. Figure 3 provides valuable data on the locomotor effects of A13 region photoactivation in both control and 6-OHDA mice. However, a more detailed analysis of the changes in locomotion during stimulation would enhance our understanding of the pro-locomotor effects, especially in the context of 6-OHDA lesions. For example, it would be informative to explore whether the probability of locomotion changes during stimulation in the control and 6-OHDA groups. Investigating reaction time, speed, total distance, and could reveal how A13 is influencing locomotion, particularly after 6-OHDA lesions. The laboratory of Whelan has a deep knowledge of locomotion and the neural circuits driving it so these features may be instructive to infer insights on the neural circuits driving movement. On the same line, examining features like the frequency or power of stimulation related to walking patterns may help elucidate whether A13 is engaging with the Mesencephalic Locomotor Region (MLR) to drive the pro-locomotor effects. These insights would provide a more comprehensive understanding of the mechanisms underlying A13-mediated locomotor changes in both healthy and pathological conditions.*

Thank you for these suggestions. We have reorganized Figure 3 to highlight the metrics by separating the 6-OHDA from the Sham experiments (3F-J, which highlights distance travelled, average speed and duration). We have also added additional text to highlight these metrics better in the text. We have relabelled Supplementary Figure S3, which presents reaction time as latency to initiate locomotion and updated the main text to address the reviewers' points.

> ***Reviewer #2 (Public Review):***
>
> *Summary:*

> *The paper by Kim et al. investigates the potential of stimulating the dopaminergic A13 region to promote locomotor restoration in a Parkinson's mouse model. Using wild-type mice, 6-OHDA injection depletes dopaminergic neurons in the substantia nigra pars compacta, without impairing those of the A13 region and the ventral tegmentum area, as previously reported in humans. Moreover, photostimulation of presumably excitatory (CAMKIIa) neurons in the vicinity of the A13 region improves bradykinesia and akinetic symptoms after 6-OHDA injection. Whole-brain imaging with retrograde and anterograde tracers reveals that the A13 region undergoes substantial changes in the distribution of its afferents and projections after 6-OHDA injection. The study suggests that if the remodeling of the A13 region connectome does not promote recovery following chronic dopaminergic depletion, photostimulation of the A13 region restores locomotor functions.*
>
> *Strengths:*
>
> *Photostimulation of presumably excitatory (CAMKIIa) neurons in the vicinity of the A13 region promotes locomotion and locomotor recovery of wild-type mice 1 month after 6-OHDA injection in the medial forebrain bundle, thus identifying a new potential target for restoring motor functions in Parkinson's disease patients.*
>
> *Weaknesses:*
>
> *Electrical stimulation of the medial Zona Incerta, in which the A13 region is located, has been previously reported to promote locomotion (Grossman et al., 1958). Recent mouse studies have shown that if optogenetic or chemogenetic stimulation of GABAergic neurons of the Zona Incerta promotes and restores locomotor functions after 6-OHDA injection (Chen et al., 2023), stimulation of glutamatergic ZI neurons worsens motor symptoms after 6-OHDA (Lie et al., 2022).*

Thank you - we have added this reference. It is helpful as Grossman did stimulate the zona incerta in the cat and elicit locomotion, suggesting that stimulation of the area in normal mice has external validity. Grossman’s results prompted a later clinical examination of the zona incerta, but it concentrated on the zona incerta regions close to the subthalamic regions (Ossowska 2019), further caudal to the area we focused on. Chen et al. (2023) targeted the area in the lateral aspect of central/medial zona incerta, formed by dorsal and ventral zona incerta, which may account for the differing results. Our data were robust for stimulation of the medial aspect of the rostromedial zona incerta. The thigmotactic behaviour that we observed in our work that focused on CamKII neurons has not been observed with chemogenetic, optogenetic activation or with photoinhibition of GABAergic central/medial ZI (Chen et al. 2023).

GABAergic activation of mZI to Cuneiform projections (Sharma et al. 2024) also did not produce thigmotactic behavior. We have added these points to the discussion.

> *Although CAMKIIa is a marker of presumably excitatory neurons and can be used as an alternative marker of dopaminergic neurons, behavioral results of this study raise questions about the neuronal population targeted in the vicinity of the A13 region. Moreover, if YFP and CHR2-YFP neurons express dopamine (TH) within the A13 region (Fig. 2), there is also a large population of transduced neurons within and outside of the A13 region that do not, thus suggesting the recruitment of other neuronal cell types that could be GABAergic or glutamatergic.*

We found that CamKII transfection of the A13 region was extremely effective in promoting locomotor activity, which was critical for our work in exploring its possible therapeutic potential. We have since quantified the cell number, we found that the c-fos cell number was increased following ChR2 activation. There is evidence of TH activation - but the data suggest that other cell types contribute. C-fos alone is a blunt tool to assess specificity - rather, it is better at showing overall photostimulus efficacy - which we have demonstrated. Moreover, there is evidence that cell types are not purely dopaminergic, with GABA co-localized (Negishi et al. 2020). We acknowledge that specific viral approaches that target the GABAergic, glutamatergic, and dopaminergic circuits would be very useful. The range of tools to target A13 dopaminergic circuits is more limited than the SNc, for example, because the A13 region lacks DAT, and TH-IRES-Cre approaches, while helpful, are less specific than DAT-Cre mouse models. Intersectional approaches targeting multiple transmitters (glutamate & dopamine, for example) may be one solution as we do not expect that a single transmitter-specific pathway would work, as well as broad targeting of the A13 region. Our recent work suggests that GABAergic neuron activation may have more general effects on behaviour rather than control of ongoing locomotor parameters (Sharma et al. 2024). Recent work shows a positive valence effect of dopamine A13 activation on motivated food-seeking behavior, which differs from consummatory behavior observed with GABAergic modulation (Ye, Nunez, and Zhang 2023). Chemogenetic inactivation and ablation of dopaminergic A13 revealed that they contribute to grip strength and prehensile movements, uncoupling food-seeking grasping behavior from motivational factors (Garau et al. 2023). Overall, this suggests differing effects of GABA compared to DA and/or glutamatergic cell types, consistent with our effects of stimulating CamKII. The discussion has been updated.

> *Regarding the analysis of interregional connectivity of the A13 region, there is a lack of specificity (the viral approach did not specifically target the A13 region), the number of mice is low for such correlation analyses (2 sham and 3 6-OHDA mice), and there are no statistics comparing 6-OHDA versus sham (Fig. 4) or contra- versus ipsilesional sides (Fig. 5). Moreover, the data are too processed, and the color matrices (Fig. 4) are too packed in the current format to enable proper visualization of the data. The A13 afferents/efferents analysis is based on normalized relative values; absolute values should also be presented to support the claim about their upregulation or downregulation.*

Generally, papers using tissue-clearing imaging approaches have low sample sizes due to technical complexity and challenges. The technical challenges of obtaining these data were substantial in both collection and analysis. There are multiple technical complexities arising from dual injections (A13 and MFB coordinates) and targeting the area correctly. The A13 region is difficult to target as it spans only around 300 μm in the anterior-posterior axis. While clearing the brain takes weeks, and light-sheet imaging also takes time, the time necessary to analyze the tissue using whole-brain quantification is labor intensive, especially with a lack of a standardized analysis pipeline from atlas registrations, signal segmentations, and quantifications. The field is still relatively new, requiring additional time to refine pipelines.

Correlation matrices are often used in analyzing connectivity patterns on a brain-wide scale, as they can identify any observable patterns within a large amount of data. We used correlation matrices to display estimated correlation coefficients between the afferent and efferent proportions from one brain subregion to another across 251 brain regions in total in a pairwise manner (not for hypothesis testing). We provided descriptive statistics (mean and error bars) in the original Figure 5C and G. As mentioned in comments for Reviewer 1, we have now presented the data in revised Figure 4 and 5 that focuses specifically on motor-related pathways to provide information on possible pathways. The has simplified the correlation matrices and highlighted the differences in 6-OHDA efferent data especially. As suggested, raw values are shared in a supplemental file on our data repository.

> *In the absence of changes in the number of dopaminergic A13 neurons after 6-OHDA injection, results from this correlation analysis are difficult to interpret as they might reflect changes from various impaired brain regions independently of the A13 region.*

We acknowledge that models of Parkinson's disease, particularly those using 6-OHDA, induce plasticity in various regions, which may subsequently affect A13 connectivity. We aim to emphasize the residual, intact A13 pathways that could serve as therapeutic targets in future investigations. This emphasis is pertinent in the context of potential clinical applications, as the overall input and output to the region fundamentally dictate the significance of the A13 region in lesioned nigrostriatal models. We agree with the reviewer that the changes certainly can be independent of A13; however, the fact that there was a significant change in the connectome post-6-OHDA injection and striatonigral degeneration is in and of itself important to document. We have added a sentence acknowledging this limitation to the discussion.

> *There is no causal link between anatomical and behavioral data, which raises questions about the relevance of the anatomical data.*

This point was also addressed earlier in response to a comment from Reviewer 1. Focusing on specific motor pathways is one avenue to explore. However, given that the zona incerta acts as an integrative hub, we believed it is prudent to initially examine both afferent and efferent pathways using a brain-wide approach. For instance, without employing this methodology, the potential significance of cortical interconnectivity to the A13 region might not have been fully appreciated. As mentioned previously, we will place additional emphasis on motor-related regions in our revised paper, thereby enhancing the relevance of the anatomical data presented. With these modifications, we anticipate that our data will underscore specific motor-related targets for future exploration, employing optogenetic targeting to assess necessity and sufficiency.

> *Overall, the study does not take advantage of genetic tools accessible in the mouse to address the direct or indirect behavioral and anatomical contributions of the A13 region to motor control and recovery after 6-OHDA injection.*

Our study has not specifically targeted neurons that express dopaminergic, glutamatergic, or GABAergic properties (refer to earlier comment for more detail). However, like others, we find that targeting one neuronal population often does not result in a pure transmitter phenotype. For instance, evidence suggests co-localization of dopamine neurons with a subpopulation of GABA neurons in the A13/medial zona incerta (Negishi et al. 2020). In the hypothalamus, research by Deisseroth and colleagues (Romanov et al. 2017) indicates the presence of multiple classes of dopamine cells, each containing different ratios of co-localized peptides and/or fast neurotransmitters. Consequently, we believe our work lays the foundation for the investigations suggested by the reviewer. Furthermore, if one considers this work in the context of a preclinical study to determine whether the A13 might be a target in human Parkinson's disease, the existing technology that could be utilized is deep brain stimulation (DBS) or electrical modulation, which would also affect different neuronal populations in a non-specific manner.

While optogenetic stimulation therapy is longer term, using CamKII combined with the DJ hybrid AAV could be a translatable strategy for targeting A13 neuronal populations in non-human primates (Watakabe et al. 2015; Watanabe et al. 2020). We have added to the discussion.

> ***Reviewer #3 (Public Review):***
>
> *Kim, Lognon et al. present an important finding on pro-locomotor effects of optogenetic activation of the A13 region, which they identify as a dopamine-containing area of the medial zona incerta that undergoes profound remodeling in terms of afferent and efferent connectivity after administration of 6-OHDA to the MFB. The authors claim to address a model of PD-related gait dysfunction, a contentious problem that can be difficult to treat with dopaminergic medication or DBS in conventional targets. They make use of an impressive array of technologies to gain insight into the role of A13 remodeling in the 6-OHDA model of PD. The evidence provided is solid and the paper is well written, but there are several general issues that reduce the value of the paper in its current form, and a number of specific, more minor ones. Also, some suggestions, that may improve the paper compared to its recent form, come to mind.*

Thank you for the suggestions and careful consideration of our work - it is appreciated.

> *The most fundamental issue that needs to be addressed is the relation of the structural to the behavioral findings. It would be very interesting to see whether the structural heterogeneity in afferent/effects projections induced by 6-OHDA is related to the degree of symptom severity and motor improvement during A13 stimulation.*

As mentioned in comments for Reviewer 1, we have performed additional analysis and present this in Figure 5. We have also revised Figure 4, focusing on motor regions. Our work will provide a roadmap for future studies to disentangle divergent or convergent A13 pathways that are involved in different or all PD-related motor symptoms. Because we could not measure behavioural change in the same animals studied with the anatomic study (essentially because the optrode would have significantly disrupted the connectome we are measuring), we cannot directly compare behaviour to structure.

> *The authors provide extensive interrogation of large-scale changes in the organization of the A13 region afferent and efferent distributions. It remains unclear how many animals were included to produce Fig 4 and 5. Fig S5 suggests that only 3 animals were used, is that correct? Please provide details about the heterogeneity between animals. Please provide a table detailing how many animals were used for which experiment. Were the same animals used for several experiments?*

The behavioral set and the anatomical set were necessarily distinct. In the anatomical experiments, we employed both anterograde and retrograde viral approaches to target the afferent and efferent A13 populations with fluorescent proteins. For the behavioral approach, a single ChR2 opsin was utilized to photostimulate the A13 region; hence combining the two populations was not feasible. We were also concerned that the optrode itself would interfere with connectomics. A lower number of animals were used for the whole-brain work due to technical limitations described earlier. We have now provided additional information regarding numbers in all figures and the text. Using Spearman's correlation analysis, we found afferent and efferent proportions across animals to be consistent, with an average correlation of 0.91, which is reported in Figure S6.

> *While the authors provide evidence that photoactivation of the A13 is sufficient in driving locomotion in the OFT, this pro-locomotor effect seems to be independent of 6-OHDA-induced pathophysiology. Only in the pole test do they find that there seems to be a difference between Sham vs 6-OHDA concerning the effects of photoactivation of the A13. Because of these behavioral findings, optogenic activation of A13 may represent a gain of function rather than disease-specific rescue. This needs to be highlighted more explicitly in the title, abstract, and conclusion.*

Optogenetic activation of A13 may represent a gain of function in both healthy and 6-OHDA mice, highlighting a parallel descending motor pathway that remains intact. 6-OHDA lesions have multiple effects on motor and cognitive function. This makes a single pathway unlikely to rescue all deficits observed in 6-OHDA models. The lack of locomotion observed in 6-OHDA models can be reversed by A13 region photostimulation. Therefore, this is a reversal of a loss of function, in this case. However, the increase in turning represents a gain of function. We have highlighted this as suggested in the discussion.

> *The authors claim that A13 may be a possible target for DBS to treat gait dysfunction. However, the experimental evidence provided (in particular the lack of disease-specific changes in the OFT) seems insufficient to draw such conclusions. It needs to be highlighted that optogenetic activation does not necessarily have the same effects as DBS (see the recent review from Neumann et al. in Brain: https://pubmed.ncbi.nlm.nih.gov/37450573/). This is important because ZI-DBS so far had very mixed clinical effects. The authors should provide plausible reasons for these discrepancies. Is cell-specificity, which only optogenetic interventions can achieve, necessary? Can new forms of cyclic burst DBS achieve similar specificity (Spix et al, Science 2021)? Please comment.*

Thank you for the valuable comments. They have been incorporated into the discussion.

Our study highlights a parallel motor pathway provided by the A13 region that remains intact in 6-OHDA mice and can be sufficiently driven to rescue the hypolocomotor pathology observed in the OFT and overcome bradykinesia and akinesia. The photoactivation of ipsilesional A13 also has an overall additive effect on ipsiversive circling, representing a gain of function on the intact side that contributes to the magnitude of overall motor asymmetry against the lesioned side. The effects of DBS are rather complex, ranging from micro-, meso-, to macro-scales, involving activation, inhibition, and informational lesioning, and network interactions. This could contribute to the mixed clinical effects observed with ZI-DBS, in addition to differences in targeting and DBS programming among the studies (see review (Ossowska 2019)). Also the DBS studies targeting ZI have never targeted the rostromedial ZI which extends towards the hypothalamus and contains the A13. Furthermore, DBS and electrical stimulation of neural tissue, in general, are always limited by current spread and lower thresholds of activation of axons (e.g., axons of passage), both of which can reduce the specificity of the true therapeutic target. Optogenetic studies have provided mechanistic insights that could be leveraged in overcoming some of the limitations in targeting with conventional DBS approaches. Spix et al. (2021) provided an interesting approach highlighting these advancements. They devised burst stimulation to facilitate population-specific neuromodulation within the external globus pallidus. Moreover, they found a complementary role for optogenetics in exploring the pathway-specific activation of neurons activated by DBS. To ascertain whether A13 DBS may be a viable therapy for PD gait, it will be necessary to perform many more preclinical experiments, and tuning of DBS parameters could be facilitated by optogenetic stimulation in these murine models. We have added to the discussion.

> *In a recent study, Jeon et al (Topographic connectivity and cellular profiling reveal detailed input pathways and functionally distinct cell types in the subthalamic nucleus, 2022, Cell Reports) provided evidence on the topographically graded organization of STN afferents and McElvain et al. (Specific populations of basal ganglia output neurons target distinct brain stem areas while collateralizing throughout the diencephalon, 2021, Neuron) have shown similar topographical resolution for SNr efferents. Can a similar topographical organization of efferents and afferents be derived for the A13/ ZI in total?*

The ZI can be subdivided into four subregions in the antero-posterior axis: rostral (ZIr), dorsal (ZId), ventral (ZIv), and caudal (ZIc) regions. The dorsal and ventral ZI is also referred together as central/medial/intermediate ZI. There are topographical gradients in different cell types and connectivity across these subregions (see reviews: (Mitrofanis 2005; Monosov et al. 2022; Ossowska 2019). Recent work by Yang and colleagues (2022) demonstrated a topographical organization among the inputs and outputs of GABAergic (VGAT) populations across four ZI subregions. Given that A13 region encompasses a smaller portion (the medial aspect) of both rostral and medial/central ZI (three of four ZI subregions) and coexpress VGAT, A13 region likely falls under rostral and intermediate medial ZI dataset found in Yang et al. (2022). With our data, we would not be able to capture the breadth of topographical organization shown in Yang et al (2022).

In conclusion, this is an interesting study that can be improved by taking into consideration the points mentioned above.

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations For The Authors):***
>
> *(1) Figure 2 indeed presents valuable information regarding the effects of A13 region photoactivation. To enhance the comprehensiveness of this figure and gain a deeper understanding of the neurons driving the pro-locomotor effect of stimulation, it would be beneficial to include quantifications of various cell types:*
>
> *• cFos-Positive Cells/TH-Positive Cells: it can help determine the impact of A13 stimulation on dopaminergic neurons and the associated pro-locomotor effect in the healthy condition and especially in the context of Parkinson's disease (PD) modeling.*
>
> *• cFos-Positive Cells /TH-Negative Cells: Investigating the number of TH-negative cells activated by stimulation is also important, as it may reveal non-dopaminergic neurons that play a role in locomotor responses. Identifying the location and characteristics of these TH-negative cells can provide insights into their functional significance.*

We have completed this analysis. The data is presented in Figure 2F, where we show increased c-fos intensity with photoactivation. We observed an increase in the number of cells activated in the A13 region. However, we did not definitively see increases in TH+ cells, suggesting a heterogeneous set of neurons responsible for the effects—possibly glutamatergic neurons.

> *Incorporating these quantifications into Figure 2 would enhance the figure's informativeness and provide a more comprehensive view of the neuronal populations involved in the locomotor effects of A13 stimulation.*

We have added text and a new graph.

> *(2) Refer to Figure 3. In the main text (page 5) when describing the animal with 6-OHDA the wrong panels are indicated. It is indicated in Figure 2A-E but it should be replaced with 3A-E.*
>
> *Please do that.*

Done, and we have updated the figure to improve readability, by separating the 6-OHDA findings from sham in all graphs.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *Abstract*

> *Page 1: Inhibitory or lesion studies will be necessary to support the claim that the global remodeling of afferent and efferent projections of the A13 region highlights the Zona Incerta's role as a crucial hub for the rapid selection of motor function.*

Overall, there is quite a bit of evidence that the zona incerta is a hub for afferent/efferents.

Mitrofanis (2005) and, more recently, Wang et al. (2020) summarize some of the evidence. Yang (2022) illustrates that the zona incerta shows multiple inputs to GABAergic neurons and outputs to diverse regions. Recent work suggests that the zona incerta contributes to various motor functions such as hunting, exploratory locomotion, and integrating multiple modalities (Zhao et al. 2019; Wang et al. 2019; Monosov et al. 2022; Chometton et al. 2017). The introduction has been updated.

> *Introduction*
>
> *Page 2, paragraph 2: "However, little attention has been placed on the medial zona incerta (mZI), particularly the A13, the only dopamine-containing region of the rostral ZI" Is the A13 region located in the rostral or medial ZI or both?*

It should have been written “rostromedial” ZI. The A13 is located in the medial aspect of rostromedial ZI. Introduction has been updated.

> *Page 2, para 3: Li et al (2021) used a mini-endoscope to record the GCaMP6 signal. Masini and Kiehn, 2022 transiently blocked the dopaminergic transmission; they never used 6-OHDA.*
>
> *Please correct through the text.*

Corrected.

> *Page 2, para 4: the A13 connectome encompasses the cerebral cortex,... MLR. The MLR is a functional region, correct this for the CNF and PPN.*

Corrected.

> *Page 3, the last paragraph of the introduction could be clarified by presenting the behavioral data first, followed by the anatomy.*

This has been corrected

> *Figure 1 is nice and clear, and well summarizes the experimental design.*

Thank you.

> *Figure 2 shows an example of the extent of the ChR2-YFP expression and the position of an optical fiber tip above the dopaminergic A13 region from a mouse. Without any quantification, these images could be included in Figure 1. Despite a very small volume (36.8nL) of AAV, the extent of ChR2-YFP expression is quite large and includes dopaminergic and unidentified neurons within the A13 region but also a large population of unidentified neurons outside of it, thus raising questions about the volume and the types of neurons recruited.*

This is an important consideration. The issue of viral spread is complex and depends on factors including tissue type, serotype, and promotor of the virus. Li et al. (2021), for example, used different virus serotypes and promotors, injecting 150nL, whereas we used AAV DJ, injecting 36.8nL. AAV-DJ is a hybrid viral type consisting of multiple serotypes. It has a high transduction efficiency, which leads to greater gene delivery than single-serotype AAV viral constructs (Mao et al. 2016). A secondary consideration regarding translation was that AAV-DJ could effectively transduce non-primate neurons (Watanabe et al. 2020). We have addressed the issue of neurons recruited earlier, provided c-Fos quantification, and provided a new supplementary figure showing viral spread (Figure S1).

> *Anatomical reconstruction of the extent of the ChR2-YFP expression and the location of the tip of the optical fiber will be necessary to confirm that ChR2-YFP expression was restricted to the A13 region.*

We will provide additional information regarding viral spread, ferrule tip placement, and c-fos cell counts. This has been done in Figure 2 and we also present a new Figure S1 where we have quantified the viral spread.

> *Page 5, 1st para: Double-check the references, as not all of them are 6-OHDA injections in the MLF.*

Corrected. Removed Kiehn reference.

> *Page 5, 1st para, 4th line: Replace ferrule with optical canula or fiber.*

Done

> *Page 5, 1st para, 9th line: Replace Figure 2 with Figure 3.*

Done

> *Page 5, 2nd para: About the refractory decrease in traveled distance by sham-ChR2 mice: is this significant?*

It was not significant (Figure S1C, 1-way RM ANOVA: $F_{5,25} = 0.486$, $P = 0.783$). This has been updated in the text.

> *Figure 3 showing behavioral assessments is nice, but the stats are not always clear. In Fig 3A, are each of the off and on boxes 1 minute long? The figure legend states the test lasts 1 min, but isn't it 4 minutes? In Figure 3B-E and 3J-M, what are the differences? Do the stats identify a significant difference only during the stimulation phase? Fig. 3F-I are nice and could have been presented as primary examples prior to data analysis in Fig. 3B-E. Group labels above the graph would help.*

Yes, the off-on boxes are 1 minute long. The error is corrected in the legend. Great suggestion for F-I - they have been moved ahead of the summary figures. We have also updated new Fig 3F-,I, J, L, M) to make the differences between 6-OHDA and sham graphs easier to visualize. The stats do indicate a significant difference during the stimulation phase. We have added group labels, and reorganized the figure, and it is much easier to read now.

> *Fig. 3L-M, what do PreSur, Post, and Ferrule mean? I assume that Ferrule refers to mice tested with the optical fiber without stimulation, whereas Stim. refers to the stimulation. It would be helpful to standardize the format of stats in Fig. 3B-E and 3-J-M. What are time points a, b, and c referring to?*

We have renamed the figure names to be more intuitive. We have standardized the presentation of statistics in the figure, and eliminated the a,b,c nomenclature. We have also updated the caption to provide descriptions of the tests in Fig 3 L-M.

> *Figure S2A: the higher variability in 6-OHDA-YFP mice in comparison to 6-OHDA-ChR2 mice prior to stimulation suggests that 6-OHDA-YFP mice were less impaired. Why use boxplots only for these data? Would a pairwise comparison be more appropriate?*

We have removed these plots from Figure S2. We now present the Baseline to Pre values across the experimental timespan to illustrate the fact that distance travelled returned to baseline values for all trials conducted.

> *Fig. S2B: add the statistical marker.*

We have removed this from Figure S2.

> *Page 7, para 1, line 8: to add "in comparison to 6-OHDA-YFP and YFP mice" to during photostimulation... (Figure 3E).*

Done

> *Page 7, para 3, line 5: about larger improvement, replace "sham ChR2" with "6-OHDA."*

Done

> *Page 8, para 1, line 4: Perier et al., 2000 reported that 6-OHDA injection increased the firing frequency of the ZI over a month.*

Added the timeframe to this sentence.

> *Page 8, para 2, line 1: Since the results were expected, add some references.*

Done.

> *Page 8, para 3, line 4. Double-check the reference.*

Corrected.

> *Page 8: About large-scale changes in the A13 region, the relevance of correlation matrices is difficult to grasp. Analysis of local connectivity would have been more informative in the context of GABAergic and glutamatergic neurons of the ZI in the vicinity of the A13 region.*

We have updated the figures for connectivity throughout the manuscript. Overall, there are new Figures 4 and 5 in the main text. We also provide a revised Supplementary Figure 8. Unfortunately, we could not do that experiment regarding local connectivity. In light of our new work (Sharma et al. 2024), it is clear that this will be critical going forward.

> *Page 8, para 3, line: given Fig. 2, there is concern about the claim that only the A13 region was targeted. The time of the analysis after 6-OHDA should be mentioned. Some sections of the paragraph could be moved to methods.*

We have provided more information about the viral spread in the text and Supplementary Figure 1. The functional and anatomical experiments are separate, which we realize caused confusion. We have mentioned analysis time after 6-OHDA and inserted this into the text.

> *Fig. 4: The color code helps the reader visualize distribution differences. However, statistical analyses comparing 6-OHDA versus sham should be included. Quantification per region would greatly help readers visualize the data and support the conclusion. The relationship between the type of correlation (positive or negative) and absolute change (increase or decrease) is unknown in the current format, which limits the interpretation of the data. Moreover, examples of raw images of axons and cells should be presented for several brain regions. The experimental design with a timeline, as in Fig. 1, would be helpful. The legend for Fig. 4 is a bit long. Some sections are very descriptive, whereas others are more interpretive.*

We have provided a new Figure 5 where we present quantification per region, and the correlation matrices have been updated in Figure 4. We have also focused on motor regions as mentioned earlier. We also provide examples of raw regions in Supplementary Figure 8. Raw values are shared on our data repository.

> *Page 10, para 1, line 1: add "afferent" to "changes in -afferent and- projection patterns."*

Done

> *Page 10, para 1, line 9: remove the 2nd "compared to sham" in the sentence.*

Done

> *Page 10, para 1, line 10: remove "coordinated" in "several regions showed a coordinated reduction in afferent density." We cannot say anything about the timing of events, as there is only info at 1 month.*

Done

> *Page 10, para 2: the section should be written in the past tense.*

Done

> *Page 13, para 2, the last sentence is overstated. Please remove "cells" and refer to the A13 region instead.*

Done

> *About differential remodelling of the A13 region connectome: Figure 5C and 5G: The proportion of total afferents ipsi- and contralateral to 6-OHDA injection argues that the A13 region primarily receives inputs from the cortical plate and the striatum. Unfortunately, there are no statistics.*

Due to the small sample size, we provided descriptive statistics (mean and error bars) in Figure 5A. As mentioned in comments for Reviewers 1 and 2, we have revised Figure 5 to present data focusing on motor-related pathways to provide clarity. In addition, absolute values are shared on our data repository.

> *Figure 5 D and 5H: Changes in the proportion of total afferents/projections are relatively modest (less than 10% of the whole population for the highest changes). There is no standard deviation for these data and no statistics. Do they reflect real changes or variability from the injection site?*

The changes are relatively modest (less than 10%) since a small brain region usually provides a small proportion of total input (McElvain et al. 2021; Yang et al. 2022). The changes in the proportions reflect real differences between average proportions observed in sham and 6-OHDA mice. The variability in the total labelling of neurons and fibers was minimized by normalizing individual regional counts against total counts found in each animal. This figure has been updated as reviewers requested.

> *Fig 5F and H: The example in F shows a huge decrease in the striatum, but H indicates only a 2% change, which makes the example not very representative. Absolute values would be helpful.*

While a 2% change may seem small, it represents a relatively large change in the A13 efferent connectome. To provide further clarity, we have provided absolute values as suggested in our new supplemental table.

> *Figure 6 is inaccurate and unnecessary.*

Figure 6 has been removed.

> *Discussion*
>
> *Although interesting, the discussion is too long.*

The discussion has been reduced by about three quarters of a page.

> *Methods*
>
> *Page 17, para 1: include the stereotaxic coordinates of the optical cannula above the A13 region.*

Added.

References

Chen, Fenghua, Junliang Qian, Zhongkai Cao, Ang Li, Juntao Cui, Limin Shi, and Junxia Xie. 2023. "Chemogenetic and Optogenetic Stimulation of Zona Incerta GABAergic Neurons Ameliorates Motor Impairment in Parkinson's Disease." i Science 26 (7). https://doi.org/ 10.1016/j.isci.2023.107149.

Chometton, S., K. Charrière, L. Bayer, C. Houdayer, G. Franchi, F. Poncet, D. Fellmann, and P. Y. Risold. 2017. "The Rostromedial Zona Incerta Is Involved in Attentional Processes While Adjacent LHA Responds to Arousal: C-Fos and Anatomical Evidence." Brain Structure & Function 222 (6): 2507–25.

Garau, Celia, Jessica Hayes, Giulia Chiacchierini, James E. McCutcheon, and John Apergis-Schoute. 2023. "Involvement of A13 Dopaminergic Neurons in Prehensile Movements but Not Reward in the Rat." Current Biology: CB, October.

https://doi.org/ 10.1016/j.cub.2023.09.044.

Li, Zhuoliang, Giorgio Rizzi, and Kelly R. Tan. 2021. “Zona Incerta Subpopulations Differentially Encode and Modulate Anxiety.” Science Advances 7 (37): eabf6709.

Mao, Yingying, Xuejun Wang, Renhe Yan, Wei Hu, Andrew Li, Shengqi Wang, and Hongwei Li. 2016. “Single Point Mutation in Adeno-Associated Viral Vectors -DJ Capsid Leads to Improvement for Gene Delivery in Vivo.” BMC Biotechnology 16 (January):1.

McElvain, Lauren E., Yuncong Chen, Jeffrey D. Moore, G. Stefano Brigidi, Brenda L. Bloodgood, Byung Kook Lim, Rui M. Costa, and David Kleinfeld. 2021. “Specific Populations of Basal Ganglia Output Neurons Target Distinct Brain Stem Areas While Collateralizing throughout the Diencephalon.” Neuron 109 (10): 1721–38.e4.

Mitrofanis, J. 2005. “Some Certainty for the ‘Zone of Uncertainty’? Exploring the Function of the Zona Incerta.” Neuroscience 130 (1): 1–15.

Monosov, Ilya E., Takaya Ogasawara, Suzanne N. Haber, J. Alexander Heimel, and Mehran Ahmadlou. 2022. “The Zona Incerta in Control of Novelty Seeking and Investigation across Species.” Current Opinion in Neurobiology 77 (December):102650.

Negishi, Kenichiro, Mikayla A. Payant, Kayla S. Schumacker, Gabor Wittmann, Rebecca M. Butler, Ronald M. Lechan, Harry W. M. Steinbusch, Arshad M. Khan, and Melissa J. Chee. 2020. “Distributions of Hypothalamic Neuron Populations Coexpressing Tyrosine Hydroxylase and the Vesicular GABA Transporter in the Mouse.” The Journal of Comparative Neurology 528 (11): 1833–55.

Ossowska, Krystyna. 2019. “Zona Incerta as a Therapeutic Target in Parkinson’s Disease.” Journal of Neurology. https://doi.org/ 10.1007/s00415-019-09486-8.

Romanov, Roman A., Amit Zeisel, Joanne Bakker, Fatima Girach, Arash Hellysaz, Raju Tomer, Alán Alpár, et al. 2017. “Molecular Interrogation of Hypothalamic Organization Reveals Distinct Dopamine Neuronal Subtypes.” Nature Neuroscience 20 (2): 176–88.

Sharma, Sandeep, Cecilia A. Badenhorst, Donovan M. Ashby, Stephanie A. Di Vito, Michelle A. Tran, Zahra Ghavasieh, Gurleen K. Grewal, Cole R. Belway, Alexander McGirr, and Patrick J. Whelan. 2024. “Inhibitory Medial Zona Incerta Pathway Drives Exploratory Behavior by Inhibiting Glutamatergic Cuneiform Neurons.” Nature Communications 15 (1): 1160.

Spix, Teresa A., Shruti Nanivadekar, Noelle Toong, Irene M. Kaplow, Brian R. Isett, Yazel Goksen, Andreas R. Pfenning, and Aryn H. Gittis. 2021. “Population-Specific Neuromodulation Prolongs Therapeutic Benefits of Deep Brain Stimulation.” Science 374 (6564): 201–6.

Wang, Xiyue, Xiaolin Chou, Bo Peng, Li Shen, Junxiang J. Huang, Li I. Zhang, and Huizhong W. Tao. 2019. “A Cross-Modality Enhancement of Defensive Flight via Parvalbumin Neurons in Zona Incerta.” eLife 8 (April). https://doi.org/ 10.7554/eLife.42728.

Wang, Xiyue, Xiao-Lin Chou, Li I. Zhang, and Huizhong Whit Tao. 2020. “Zona Incerta: An Integrative Node for Global Behavioral Modulation.” Trends in Neurosciences 43 (2): 82–87.

Watakabe, Akiya, Masanari Ohtsuka, Masaharu Kinoshita, Masafumi Takaji, Kaoru Isa, Hiroaki Mizukami, Keiya Ozawa, Tadashi Isa, and Tetsuo Yamamori. 2015. “Comparative Analyses of Adeno-Associated Viral Vector Serotypes 1, 2, 5, 8 and 9 in Marmoset, Mouse and Macaque Cerebral Cortex.” Neuroscience Research 93 (April):144–57.

Watanabe, Hidenori, Hiromi Sano, Satomi Chiken, Kenta Kobayashi, Yuko Fukata, Masaki Fukata, Hajime Mushiake, and Atsushi Nambu. 2020. “Forelimb Movements Evoked by Optogenetic Stimulation of the Macaque Motor Cortex.” Nature Communications 11 (1): 3253.

Yang, Yang, Tao Jiang, Xueyan Jia, Jing Yuan, Xiangning Li, and Hui Gong. 2022. "Whole-Brain Connectome of GABAergic Neurons in the Mouse Zona Incerta." Neuroscience Bulletin 38 (11): 1315–29.

Ye, Qiying, Jeremiah Nunez, and Xiaobing Zhang. 2023. "Zona Incerta Dopamine Neurons Encode Motivational Vigor in Food Seeking." bioRxiv: The Preprint Server for Biology, June. https://doi.org/ 10.1101/2023.06.29.547060.

Zhao, Zheng-Dong, Zongming Chen, Xinkuan Xiang, Mengna Hu, Hengchang Xie, Xiaoning Jia, Fang Cai, et al. 2019. "Zona Incerta GABAergic Neurons Integrate Prey-Related Sensory Signals and Induce an Appetitive Drive to Promote Hunting." Nature Neuroscience 22 (6): 921–32.

https://doi.org/10.7554/eLife.90832.2.sa0